
Unlocking Noise-Resistant Vision: Key Architectural Secrets for Robust Models Against Gaussian Noise

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Abstract

We investigate which architectural and input-pipeline choices make vision models more robust to additive Gaussian noise and convert these empirical insights into simple, actionable design rules. Specifically, we performed extensive evaluations on 1,174 pretrained vision models, empirically identifying four consistent factors for improved robustness against Gaussian noise: larger stem kernels, resolution-dependent downsampling under a pre-resize sensor-noise protocol, average pooling, and supervised vision transformers (ViTs) rather than CLIP ViTs, which yield up to 506 rank improvements and 21.6%p accuracy gains. First, we prove that low-pass stem kernels attenuate noise with a gain that decreases quadratically with kernel size and that anti-aliased downsampling reduces noise energy roughly in proportion to the square of the downsampling factor. Second, we demonstrate that average pooling is unbiased and suppresses noise in proportion to the pooling window area, whereas max pooling incurs a positive bias that grows slowly with window size and yields a relatively higher mean-squared error and greater worst-case sensitivity. Third, we reveal and explain the vulnerability of CLIP ViTs via a pixel-space Lipschitz bound: The smaller normalization standard deviations used in CLIP preprocessing amplify worst-case sensitivity by up to 1.91 times relative to the Inception-style preprocessing common in supervised ViTs. Our results collectively disentangle robustness into interpretable modules, provide a theory that explains the observed trends, and build practical, plug-and-play guidelines for designing vision models more robust against Gaussian noise.

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1. Introduction

Vision models, implemented with deep neural networks, are now deployed across numerous fields, even in safety-critical applications ranging from medical imaging to autonomous driving. Their remarkable accuracy, however, conceals an uncomfortable fact: Performance can deteriorate when test images deviate—even slightly—from the training distribution (Hendrycks & Dietterich, 2019). Even light Gaussian noise can trigger misclassifications, and in autonomous vehicles, such brittleness can lead to life-threatening failures.

Recent studies have empirically discovered that the architectural design of deep neural networks strongly shapes their robustness to common image transformations. Specifically, Paul & Chen (2022); Bai et al. (2021); Naseer et al. (2021) observed that vision transformers (ViTs) often degrade less than previous convolutional networks, such as residual networks (ResNets), under various corruptions. Although promising results with ViTs have been reported, such studies typically treat each architecture as a whole, leaving unanswered which specific internal choices contribute to gains in robustness.

In this study, we dissect the robustness of vision models under Gaussian noise, showing that specific architectural and input-pipeline choices are key factors in determining robustness. We performed extensive experiments on available vision models from the `timm` library (Wightman, 2019), as well as controlled experiments; our empirical meta-analysis compares architectures pairwise within the vision models, which enables us to isolate key factors, thereby revealing four interesting design and pipeline patterns that improve robustness against Gaussian noise:

- **Larger** stem kernels, such as larger patch sizes in ViTs, rather than **smaller** ones,
- **Resolution-dependent downsampling** under a pre-resize sensor-noise protocol, where smaller native input resolutions induce stronger deterministic downsampling before model input,
- **Average** pooling, rather than **max** pooling, and
- **Supervised** learning ViTs, rather than **CLIP** ViTs.

Extending these empirical observations, we also derive several theoretical results that account for the differences in

these choices. Specifically, we prove that noise gain decays quadratically with the stem kernel size and that downsampling after anti-alias filtering yields analogous gains (Section 4). Then we analyze Gaussian-noise error formulas for both pooling operators, showing that average pooling is unbiased with decreased variance, whereas max pooling incurs a positive bias and a higher mean-squared error (Section 5). Finally, we demonstrate that the vulnerability of CLIP ViTs is primarily caused by the choice of mean-std normalization, whose effect is proven with Lipschitz bounds (Section 6).

2. Related Work

Robustness Literature and Positioning of This Study. Robustness to common corruptions is typically evaluated using ImageNet-C (Hendrycks & Dietterich, 2019). A consistent observation across studies is that ViTs often degrade less than CNNs do under such corruptions (Paul & Chen, 2022; Bai et al., 2021; Naseer et al., 2021). However, most prior comparisons treat architectures as monolithic families or vary training recipes, making it hard to isolate which micro-architectural choices drive robustness. Furthermore, multiple corruptions, such as brightness changes and blur, are mixed in. In contrast to these complex corruptions and architectures, we design a systematic evaluation protocol to isolate the effect of each factor. Furthermore, we select Gaussian noise due to its approximation of aggregate perturbations by the central limit theorem and its prevalence in real-world imaging, such as sensor readout and thermal noise. To this end, our experiments disentangle four architectural and input-pipeline choices across pretrained models and controlled settings, enabling clean attribution. Our findings align with prior results on robustness studies (Paul & Chen, 2022; Boureau et al., 2010) and add causal, quantitative explanations. The parts below review related work that corresponds to the empirical design patterns we identified for enhancing Gaussian noise robustness.

Anti-Aliasing, Kernels, and Resolution. Aliasing and downsampling are known to affect shift stability, generalization, and robustness in modern vision models (Zhang, 2019; Vasconcelos et al., 2021; Ribeiro & Schön, 2021; Grabinski et al., 2022; Zou et al., 2023), and analogous anti-aliasing ideas have been explored for ViTs (Qian et al., 2021). Building on this line, we provide a Gaussian-noise-specific account with explicit scaling laws and large-scale validation across 1,174 timm models: The output noise energy decays quadratically with the stem kernel size and the anti-aliased downsampling factor, explaining why larger stem kernels and pre-model downsampling associated with smaller native resolutions attenuate sensor noise.

Pooling under Additive Noise. Classical analysis shows that average pooling is unbiased with variance reduction,

whereas max pooling introduces a positive bias under Gaussian noise (Boureau et al., 2010); recent studies further clarify when max pooling aids invariance despite worse noise behavior (Matoba et al., 2023). We extend this line and empirically verify the predicted advantage of average pooling over max pooling across multiple datasets.

Normalization, CLIP Preprocessing, and Lipschitz Sensitivity. Vision models employ specific per-channel mean-std preprocessing, which, according to Lipschitz-based robustness theory (Virmaux & Scaman, 2018; Gouk et al., 2021; Tsuzuku et al., 2018), directly rescales pixel-space sensitivity. We make this connection explicit: Smaller channel standard deviations enlarge the end-to-end Lipschitz bound, predicting greater worst-case and mean-squared sensitivity to additive noise.

3. Why Gaussian Noise?

Our study focuses on robustness against additive Gaussian noise, which essentially captures comparable or even worst-case robustness against common image corruptions. Indeed, additive Gaussian noise is the least favorable among all perturbations whose covariance is spectrally bounded, which makes it a conservative measure of robustness.

Setup. Let $x \in [0, 1]^d$ be an image, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ be a vision model. A corruption produces $x' = \mathcal{C}(x, \xi)$ with $\Delta := x' - x$, and we write $\delta_f := f(x + \Delta) - f(x)$. For small perturbations, we linearize at x :

$$\begin{aligned} f(x + \Delta) &= f(x) + J_f(x)\Delta + r(x, \Delta), \\ \|r(x, \Delta)\|_2 &\leq \frac{L(x)}{2} \|\Delta\|_2^2, \end{aligned} \quad (1)$$

where $J_f(x) \in \mathbb{R}^{k \times d}$ is the Jacobian and $L(x)$ bounds the local Hessian.

Gaussian as a Least-Favorable Perturbation. We show that, under a natural variance constraint, Gaussian noise maximizes the expected feature-space mean-squared error.

Theorem 3.1 (Gaussian envelope under a variance constraint). *Let Δ be any zero-mean perturbation with covariance $\Sigma_\Delta \succeq 0$ satisfying $\Sigma_\Delta \preceq \sigma^2 I_d$. Then, under the local model Eq. 1, we have*

$$\begin{aligned} \mathbb{E}[\|f(x + \Delta) - f(x)\|_2^2] &= \text{tr}(J_f(x)\Sigma_\Delta J_f(x)^\top) + O(\mathbb{E}\|\Delta\|_2^3) \\ &\leq \sigma^2 \|J_f(x)\|_F^2 + O(\mathbb{E}\|\Delta\|_2^3). \end{aligned}$$

Moreover, if $\eta \sim \mathcal{N}(0, \sigma^2 I_d)$, then $\mathbb{E}[\|f(x + \eta) - f(x)\|_2^2] = \sigma^2 \|J_f(x)\|_F^2 + O(\mathbb{E}\|\eta\|_2^3)$, so Gaussian noise saturates this upper bound on the leading term.

Table 1. Top-1 accuracy (%) on the ImageNet-1K dataset before and after adding Gaussian noise to images. For the rank difference (RankDiff), more negative values indicate better robustness under noise. Models with large kernels and smaller native resolutions under the default protocol consistently showed improved robustness.

Pretrained Model	Size	Patch Size	Resol.	Top-1 $\rightarrow$ w/ Noise	Rank $\rightarrow$ w/ Noise	RankDiff
vit_small_patch16_224.augreg.in1k	S	16 ²	224 ²	78.84 $\rightarrow$ 59.22	885 $\rightarrow$ 547	-338
vit_small_patch16_384.augreg.in1k	S	16 ²	384 ²	81.12 $\rightarrow$ 56.59	673 $\rightarrow$ 613	-60
vit_base_patch16_224.augreg.in1k	B	16 ²	224 ²	79.15 $\rightarrow$ 62.21	862 $\rightarrow$ 487	-375
vit_base_patch16_384.augreg.in1k	B	16 ²	384 ²	81.10 $\rightarrow$ 60.23	676 $\rightarrow$ 524	-152
vit_base_patch32_224.augreg.in1k	B	32 ²	224 ²	74.90 $\rightarrow$ 58.44	1075 $\rightarrow$ 569	-506
vit_base_patch32_384.augreg.in1k	B	32 ²	384 ²	78.75 $\rightarrow$ 59.65	893 $\rightarrow$ 539	-354
vit_tiny_patch16_224.augreg.in21k.ft.in1k	T	16 ²	224 ²	75.46 $\rightarrow$ 40.34	1060 $\rightarrow$ 949	-111
vit_tiny_patch16_384.augreg.in21k.ft.in1k	T	16 ²	384 ²	78.42 $\rightarrow$ 30.50	921 $\rightarrow$ 1078	+157
vit_small_patch16_224.augreg.in21k.ft.in1k	S	16 ²	224 ²	81.39 $\rightarrow$ 62.43	644 $\rightarrow$ 479	-165
vit_small_patch16_384.augreg.in21k.ft.in1k	S	16 ²	384 ²	83.80 $\rightarrow$ 62.25	349 $\rightarrow$ 484	+135
vit_small_patch32_224.augreg.in21k.ft.in1k	S	32 ²	224 ²	76.00 $\rightarrow$ 57.14	1044 $\rightarrow$ 601	-443
vit_small_patch32_384.augreg.in21k.ft.in1k	S	32 ²	384 ²	80.48 $\rightarrow$ 57.33	740 $\rightarrow$ 596	-144
vit_base_patch8_224.augreg.in21k.ft.in1k	B	8 ²	224 ²	85.80 $\rightarrow$ 73.50	145 $\rightarrow$ 118	-27
vit_base_patch16_224.augreg.in21k.ft.in1k	B	16 ²	224 ²	84.53 $\rightarrow$ 71.19	257 $\rightarrow$ 192	-65
vit_base_patch16_384.augreg.in21k.ft.in1k	B	16 ²	384 ²	85.99 $\rightarrow$ 70.89	129 $\rightarrow$ 208	+79
vit_base_patch32_224.augreg.in21k.ft.in1k	B	32 ²	224 ²	80.71 $\rightarrow$ 65.31	719 $\rightarrow$ 392	-327
vit_base_patch32_384.augreg.in21k.ft.in1k	B	32 ²	384 ²	83.35 $\rightarrow$ 63.72	412 $\rightarrow$ 437	+25
vit_large_patch16_224.augreg.in21k.ft.in1k	L	16 ²	224 ²	85.84 $\rightarrow$ 76.62	141 $\rightarrow$ 55	-86
vit_large_patch16_384.augreg.in21k.ft.in1k	L	16 ²	384 ²	87.08 $\rightarrow$ 76.23	59 $\rightarrow$ 61	+2
vit_base_patch16_224.orig.in21k.ft.in1k	B	16 ²	224 ²	81.79 $\rightarrow$ 60.91	603 $\rightarrow$ 513	-90
vit_base_patch16_384.orig.in21k.ft.in1k	B	16 ²	384 ²	84.20 $\rightarrow$ 54.91	302 $\rightarrow$ 657	+355
vit_base_patch8_224.augreg2.in21k.ft.in1k	B	8 ²	224 ²	86.22 $\rightarrow$ 76.09	109 $\rightarrow$ 67	-42
vit_base_patch16_224.augreg2.in21k.ft.in1k	B	16 ²	224 ²	85.10 $\rightarrow$ 74.50	203 $\rightarrow$ 96	-107
vit_base_patch16_224.sam.in1k	B	16 ²	224 ²	80.24 $\rightarrow$ 57.13	771 $\rightarrow$ 602	-169
vit_base_patch32_224.sam.in1k	B	32 ²	224 ²	73.69 $\rightarrow$ 51.33	1101 $\rightarrow$ 748	-353
vit_medium_patch16_gap_256.sw.in12k.ft.in1k	M	16 ²	256 ²	84.45 $\rightarrow$ 73.07	274 $\rightarrow$ 132	-142
vit_medium_patch16_gap_384.sw.in12k.ft.in1k	M	16 ²	384 ²	85.54 $\rightarrow$ 73.98	163 $\rightarrow$ 106	-57
vit_so150m_patch16_reg4_gap_256.sbb.e250.in12k.ft.in1k	B+	16 ²	256 ²	86.68 $\rightarrow$ 77.54	81 $\rightarrow$ 38	-43
vit_so150m_patch16_reg4_gap_384.sbb.e250.in12k.ft.in1k	B+	16 ²	384 ²	87.37 $\rightarrow$ 77.30	49 $\rightarrow$ 44	-5

Connection to Other Corruptions. Other image corruptions, including noise, blur, brightness, weather, and digital artifacts, can be implemented as locally bounded operators, such as convolutions, pixelwise affine transforms, and compression. These yield perturbations $\Delta_{C,s}$ whose covariance has a bounded spectral norm $\lambda_{\max}(\text{Cov}(\Delta_{C,s})) \leq \sigma_C(s)^2$ for some effective variance level $\sigma_C(s)$. Applying Theorem 3.1 with $\sigma = \sigma_C(s)$ shows that, to leading order, the expected feature-space distortion induced by corruption C at severity s is upper-bounded by that induced by Gaussian noise $\eta \sim \mathcal{N}(0, \sigma_C(s)^2 I_d)$, which is the least favorable under the same variance budget. Furthermore, averaging over a data distribution $x \sim \mathcal{D}$ gives $\mathbb{E}_{x,\Delta} [\|f(x+\Delta) - f(x)\|_2^2] \lesssim \sigma^2 \mathbb{E}_x [\|J_f(x)\|_F^2]$, so robustness to Gaussian noise probes essentially the same Jacobian-based sensitivity that governs many common corruptions. See Appendix G and Appendix I for further discussion and empirical results.

Limitations Our study focuses exclusively on robustness to additive Gaussian noise, which, although common in imaging pipelines, does not encompass all real-world corruptions, such as adversarial perturbations, weather effects, or sensor-specific artifacts. Also, the empirical findings are derived from pretrained models in the `timm` library and controlled experiments on specific datasets, which may rep-

resent a limitation in their generalizability to other domains like medical imaging or video processing. Future work could extend these insights to broader corruptions, architectures, and datasets. See Appendix E for results on other architectures under a controlled setup.

4. Noise Attenuation by Low-Pass Kernels

ViTs have various configurations (Dosovitskiy et al., 2021), such as the size of each patch in the patch embedding and the input image size in pixels, which we refer to as the input resolution. Even within the same ViT architecture, various pretrained weights are available: They were trained with different recipes, the hyperparameter combinations used in training. For example, `vit_base_patch16_224.augreg.in1k` indicates the ViT with a model size of base, a patch size of 16 to set the size of each patch to 16×16 pixels, a resolution of 224^2 , and pretrained weights obtained using a training recipe of AugReg (Steiner et al., 2022) and a dataset of ImageNet-1K (Deng et al., 2009). Although plenty of variations in its configuration are allowed, the effect of each choice on robustness against Gaussian noise has not been clearly studied, making it difficult for practitioners to choose which one to use.

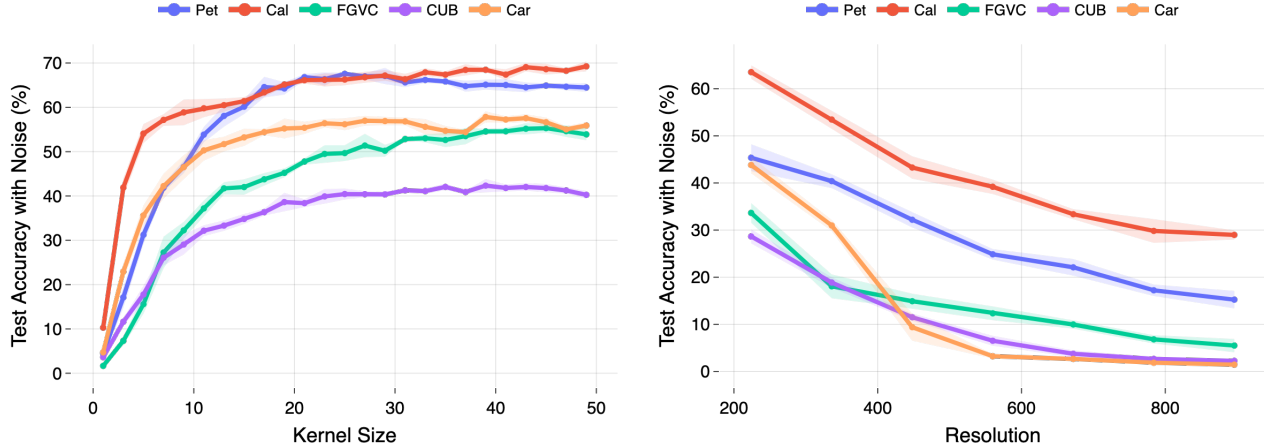


Figure 1. Classification accuracy (%) using ResNet-50 for different kernel sizes and resolutions. Larger kernels and lower evaluated resolutions under this protocol improved performance. Shaded areas represent standard deviations.

To study the effect of each architectural factor in a ViT on robustness, we performed an extensive evaluation using pre-trained ViTs with various configurations. For example, by comparing `vit_base_patch16_224.augreg_in1k` and `vit_base_patch32_224.augreg_in1k`, we can study the effect of the choice of patch sizes of 16 and 32 on performance because all other conditions remained the same. In this section, we first present empirical observations from different configurations, and then we examine the corresponding properties.

4.1. Empirical Observation

We used the `timm` library, which provides 1,174 pretrained vision models. For all pretrained models, we evaluated the top-1 accuracy (%) on the standard ImageNet-1K dataset. Then we injected Gaussian noise into the images on the ImageNet-1K dataset and measured the top-1 accuracy. We used the `Albumentations.GaussNoise()` function (Buslaev et al., 2020) with a scale factor with a range of (0.2, 0.44), where the noise was clipped to be [0, 1] and was fixed in our evaluation. In all noisy evaluations, Gaussian noise was applied before model-specific preprocessing: original image $\rightarrow$ Gaussian noise $\rightarrow$ resize $\rightarrow$ center crop $\rightarrow$ tensor conversion $\rightarrow$ model-specific mean-std normalization $\rightarrow$ model input. Although it is natural to observe a linear accuracy drop after applying a specific corruption (Recht et al., 2019; Hendrycks & Dietterich, 2019), a model with robustness would show a relatively smaller drop in top-1 accuracy. Motivated by this behavior, we identified robust models by observing relative ranking among the 1,174 models: When a model ranked 50th becomes 20th after adding Gaussian noise, we say that it demonstrates relatively stronger robustness to Gaussian noise. To investigate the model with improved rank, we computed the rank difference before and after applying Gaussian noise, where more negative values

indicate better robustness. Full rationale for rank difference and technical details are available in Appendix B and Appendix A. Full results on all models are available in the code repository at https://github.com/kmbmjn/gaussian_noise_robustness. Based on these rank differences, we compared pairs of ViTs with different configurations and investigated which architectural and input-pipeline factors contribute to improved ranking under noise.

Table 1 summarizes the top-1 accuracy and ranking changes before and after injecting Gaussian noise. We found that the rank difference was lower when a ViT had 1) a larger patch size, such as 32, and, under our default sensor-noise protocol, 2) a smaller native resolution, such as 224^2 . For example, comparing `vit_base_patch16_224.augreg_in1k` and `vit_base_patch32_224.augreg_in1k`, we observed that the model with a patch size of 32 yielded a lower rank difference than the one with a patch size of 16. We consistently observed similar behavior across multiple pretrained weights such as AugReg2, original ViTs, SAM, and others (Steiner et al., 2022; Chen et al., 2022). The same pattern appears for native resolution: models evaluated at 224^2 often had lower rank differences than matched 384^2 models. We interpret this trend as an end-to-end input-pipeline effect, because Gaussian noise is injected before resizing and smaller native resolutions apply stronger deterministic downsampling before the model input. Thus, the resolution trend should be read as evidence that downsampling in the evaluation pipeline attenuates sensor noise, rather than as a claim that lower-resolution backbones are intrinsically more robust. A matched 224-vs-384 counterfactual control in Table 9 further supports this interpretation: when noise is moved after resizing, the 384 models are favored in most cells, indicating that the default resolution trend mainly reflects pre-model downsampling

rather than intrinsic low-resolution backbone robustness. This should not be read as a monotone “smaller is always better” rule: Theorem 4.3 gives an input-noise attenuation term, but downsampling remains beneficial only while this s^{-2} gain outweighs the clean-margin loss caused by reduced image detail. Consistently, the resolution sweep in Table 10 shows that overly aggressive downsampling can lose task-relevant image information and that the beneficial regime is checkpoint- and image-dependent. These two factors were significantly more important than others, such as model size.

The patch size of a ViT corresponds to the kernel size used in the patch embedding, which is referred to as the stem. Based on these observations, we investigated whether using a larger stem kernel and lower evaluated resolutions under this protocol improves robustness to Gaussian noise on another architecture, performing controlled experiments on ResNets (He et al., 2016). Specifically, we trained ResNet-50 on five datasets, including Oxford-IIIT Pet (Parkhi et al., 2012), Caltech-101 (Fei-Fei et al., 2007), FGVC-Aircraft (Maji et al., 2013), Caltech-UCSD Birds-200-2011 (Wah et al., 2011), and Stanford Cars (Krause et al., 2013) datasets. Similar to the above ViT experiments, we trained ResNet in a standard recipe (Appendix A), obtained numerous models with different kernel sizes and resolutions, and measured classification accuracy after applying Gaussian noise.

We observed that larger kernel sizes and lower evaluated resolutions improved classification accuracy under additive Gaussian noise (Figure 1). The classification errors on noisy images tended to decrease quadratically with larger kernel sizes and stronger downsampling.

4.2. Theoretical Analysis

Now, we prove that the noise energy decays quadratically with the stem kernel size and with the anti-aliased downsampling factor associated with a resolution change. Full proofs are available in Appendix J. Throughout, $\eta \sim \mathcal{N}(0, \sigma^2 I)$ denotes independent and identically distributed (i.i.d.) Gaussian noise, and the per-pixel noise gain is the output noise energy normalized by the number of output pixels (Oppenheim, 1999).

Setup. For a kernel size $k \geq 3$, let $K_k \in \mathbb{R}^{k \times k}$ denote the linear, shift-invariant stem kernel, and $\hat{K}_k$ its DFT (Oppenheim, 1999). We consider a single, mild assumption on the stem kernel:

- (**A_{roll}**) Radial low-pass envelope at scale $1/k$: There exist $\beta, \delta > 0$ such that, for all frequencies ω ,

$$|\hat{K}_k(\omega)| \leq \phi_k(\|\omega\|), \quad \phi_k(r) := (1 + \beta kr)^{-1-\delta},$$

where ϕ_k is nonincreasing in r . This assumption works well in practical use cases (Appendix L).

Per-Pixel Noise Gain for Stem Kernel. We define

$$\begin{aligned} \gamma(k) &:= \frac{\mathbb{E}[\|K_k * \eta\|_2^2]}{\sigma^2 HW} \stackrel{(\text{Parseval})}{=} \frac{1}{HW} \sum_{\omega} |\hat{K}_k(\omega)|^2 \\ &= \|K_k\|_F^2, \end{aligned} \quad (2)$$

where H and W are the height and width. Intuitively, $\gamma(k)$ is the average squared magnitude response of the stem kernel.

Theorem 4.1 (Noise attenuation for practical low-pass stem kernel). *Assume (**A_{roll}**). Then, there exists a constant $C > 0$, independent of k , such that*

$$\gamma(k) = \frac{\mathbb{E}[\|K_k * \eta\|_2^2]}{\sigma^2 HW} \leq \frac{C}{k^2}.$$

Moreover, the k^{-2} rate is achievable.

Remark 4.2 (Practical reading of Theorem 4.1). Doubling the stem kernel size, such as the patch size from 16 to 32, quarters the output noise energy (≈ -6 dB).

Per-Output-Pixel Noise Gain for Anti-Aliased Downsampling. For a downsampling factor $s \geq 1$, we define

$$D_s := (\downarrow_s) \circ K_{g(s)}, \quad c_1 s \leq g(s) \leq c_2 s, \quad (3)$$

i.e., filter with $K_{g(s)}$ satisfying the same assumptions at scale $g(s)$ and then downsample by s . We normalize the noise gain by the number of output pixels:

$$\gamma_{\downarrow}(s) := \frac{\mathbb{E}[\|D_s \eta\|_2^2]}{\sigma^2 HW / s^2}. \quad (4)$$

Theorem 4.3 (Downsampling-driven noise attenuation). *There exists $C' > 0$ independent of s such that*

$$\gamma_{\downarrow}(s) \leq \frac{C'}{s^2}.$$

This s^{-2} rate is tight up to constants.

Remark 4.4 (Practical reading of Theorem 4.3). Resizing 384^2 to 224^2 corresponds to $s \approx 1.71$ and yields roughly $s^{-2} \approx 0.34$ of the original noise energy per output pixel (≈ -4.7 dB). In a pre-resize sensor-noise pipeline, this attenuation explains the apparent robustness advantage associated with smaller native resolutions.

5. Choice on Pooling

5.1. Empirical Observation

Extending the above analysis, we probed the effect of choosing specific architectural types of ResNets on robustness. Specifically, ResNet has several types, including ResNet-{C, D, T, S} (He et al., 2019; Wightman et al.; Guo et al., 2020),

although the effects of these choices and their underlying mechanisms on robustness have been rarely studied. Here, we trained the four ResNets on the five datasets mentioned above and compared their classification accuracy after applying Gaussian noise (Table 2).

Overall, the T and D types of ResNet demonstrated robust results against Gaussian noise, followed by the C and S types of ResNet. While there are several different factors among the four ResNets (Appendix A), the core difference is the pooling in downsampling: the T and D types of ResNet adopt average pooling with convolution in downsampling, whereas the C and S types of ResNet adopt strided 1×1 convolution in downsampling, which is equivalent to nearest-neighbor pooling followed by a 1×1 convolution.

We further explored the effect of pooling choice on robustness to Gaussian noise. Using ResNet-50, we compared the original one, which uses max pooling in the stem, and modified ResNets that adopt nearest-neighbor pooling or average pooling in the stem (Table 3). ResNets with average pooling consistently yielded robust performance against Gaussian noise among the three setups in pooling. More results for other architectures under controlled conditions are available in Appendix D.

5.2. Theoretical Analysis

We explain why average pooling is more robust than max pooling under i.i.d. additive Gaussian noise.

Setup. Consider a pooling window of size $k \geq 2$ in a single channel. Let the clean activations be $S = (S_1, \dots, S_k) \in \mathbb{R}^k$ and the observation be $S + \eta$ with i.i.d. noise $\eta \sim \mathcal{N}(0, \sigma^2 I_k)$. We define

$$X_{\text{avg}} := \frac{1}{k} \sum_{i=1}^k (S_i + \eta_i), \quad X_{\text{max}} := \max_{1 \leq i \leq k} (S_i + \eta_i),$$

their clean counterparts $S_{\text{avg}} := \frac{1}{k} \sum_i S_i$, $S_{\text{max}} := \max_i S_i$, and the errors $\delta_{\text{avg}} := X_{\text{avg}} - S_{\text{avg}}$, $\delta_{\text{max}} := X_{\text{max}} - S_{\text{max}}$. Let $\Delta := S_{(1)} - S_{(2)} \geq 0$ be the gap between the largest and second-largest entries. We also denote $T_{\text{avg}}(v) := \frac{1}{k} \sum_{i=1}^k v_i$, $T_{\text{max}}(v) := \max_{1 \leq i \leq k} v_i$, and $\|T\|_{\ell_2 \rightarrow \ell_2}$ for ℓ_2 -Lipschitz constant.

Theorem 5.1 (Average and max poolings under Gaussian noise). *For any $S \in \mathbb{R}^k$ and $\sigma > 0$, we have*

- (i) **Average pooling** is unbiased and reduces variance proportionally to the window area: $\mathbb{E}[\delta_{\text{avg}}] = 0$, $\text{Var}[\delta_{\text{avg}}] = \sigma^2/k$.
- (ii) **Max pooling** incurs a positive noise bias and admits the

following mean-squared error (MSE) controls:

$$(\text{Bias}) \quad \mathbb{E}[\delta_{\text{max}}] = \mathbb{E}[\max_i (S_i + \eta_i)] - \max_i S_i \geq 0,$$

$$(\text{Uniform-signal case}) \quad (S_1 = \dots = S_k) :$$

$$\delta_{\text{max}} = \sigma M_k, \quad \mathbb{E}[\delta_{\text{max}}^2] = \sigma^2 \mathbb{E}[M_k^2],$$

$$(\text{General case}) \quad |\delta_{\text{max}}| \leq \|\eta\|_\infty$$

$$\Rightarrow \mathbb{E}[\delta_{\text{max}}^2] \leq \sigma^2 \mathbb{E}[A_k^2],$$

where $M_k := \max_{1 \leq i \leq k} Z_i$ and $A_k := \max_{1 \leq i \leq k} |Z_i|$ with $Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. In particular, $\mathbb{E}[A_k^2] \leq 2 \log(2k) + 2$, so $\mathbb{E}[\delta_{\text{max}}^2] \leq \sigma^2 (2 \log(2k) + 2)$.

- (iii) **Adversarial worst-case sensitivity.** For any perturbation $n \in \mathbb{R}^k$, $|\frac{1}{k} \sum_i n_i| \leq \|n\|_2 / \sqrt{k}$, so $\|T_{\text{avg}}\|_{\ell_2 \rightarrow \ell_2} = k^{-1/2}$; and $|\max_i a_i - \max_i b_i| \leq \|a - b\|_\infty \leq \|a - b\|_2$, so $\|T_{\text{max}}\|_{\ell_2 \rightarrow \ell_2} \leq 1$.
- (iv) **Large-gap regime.** For $z := \Delta/\sigma$, one has $\lim_{z \rightarrow \infty} \mathbb{E}[\delta_{\text{max}}^2] = \sigma^2$; when the top index never switches under noise, max pooling is equivalent to reading a single noisy entry.

Remark 5.2 (Practical reading of Theorem 5.1). Average pooling is unbiased and cuts Gaussian noise variance by a factor k (e.g., a 2×2 window gives -6 dB). Max pooling is positively biased, and its MSE grows at most logarithmically with the window size, while also having a greater worst-case Lipschitz gain, clearly worse than average pooling.

Remark 5.3 (Average and nearest-neighbor poolings). Selecting a fixed element in the window, such as the nearest-neighbor pooling, is unbiased with an MSE σ^2 . Hence, average pooling is strictly more robust to additive Gaussian noise than nearest-neighbor pooling by a factor of k in MSE.

6. Why Are CLIP Models Vulnerable?

6.1. Empirical Observation

Although the original ViT (Dosovitskiy et al., 2021) was trained with supervised learning, the CLIP study (Radford et al., 2021) trained ViTs with contrastive learning and successfully achieved competitive performance. Currently, available pretrained weights for ViTs are largely divided into CLIP ViTs and others trained with supervised learning; we refer to the latter as supervised ViTs. The training methods and datasets differ between these two sources of ViTs, yielding different pretrained weights, while they have almost the same architecture with only a single minor difference. Nevertheless, we observed that CLIP ViTs exhibited significant vulnerabilities when Gaussian noise was applied to images (Table 4). Similar observations regarding the degraded performance of CLIP ViTs due to distribution shifts have been reported in certain studies (Shu et al., 2023; Wortsman et al., 2022); they focused on the characteristics

Table 2. Classification accuracy (%) with different choices of ResNet type. The numbers in the parentheses represent standard deviations on the five runs with different random seeds.

Dataset	Model	ResNet-50-T	ResNet-50-D	ResNet-50-C	ResNet-50-S
Configuration					
	Stem Conv	3-layer 3×3	3-layer 3×3	3-layer 3×3	3-layer 3×3
	Stem Width	24, 48, 64	32, 32, 64	32, 32, 64	64, 64, 128
	Downsample	Average	Average	Convolution	Convolution
Results					
Oxford-IIIT Pet	Val. Acc. w/ Noise	39.1 (11.1)	37.9 (9.6)	34.9 (11.4)	24.3 (3.3)
	Test Acc. w/ Noise	38.1 (11.3)	36.1 (10.3)	34.0 (10.4)	22.9 (2.4)
Caltech-101	Val. Acc. w/ Noise	62.3 (1.4)	61.2 (3.0)	58.8 (1.1)	50.9 (3.2)
	Test Acc. w/ Noise	59.7 (1.1)	59.1 (2.8)	57.8 (1.0)	49.5 (2.7)
FGVC-Aircraft	Val. Acc. w/ Noise	27.8 (1.6)	27.3 (2.4)	23.9 (1.9)	4.7 (0.9)
	Test Acc. w/ Noise	29.9 (1.1)	30.4 (1.6)	26.1 (2.1)	5.5 (0.8)
Caltech-UCSD Birds-200-2011	Val. Acc. w/ Noise	27.6 (2.0)	28.8 (0.8)	26.3 (1.3)	13.9 (0.6)
	Test Acc. w/ Noise	26.3 (2.0)	27.7 (0.6)	25.2 (1.7)	13.7 (1.1)
Stanford Cars	Val. Acc. w/ Noise	56.9 (2.3)	55.2 (2.8)	41.6 (2.3)	29.2 (1.9)
	Test Acc. w/ Noise	55.0 (1.9)	53.2 (2.7)	40.5 (2.3)	28.5 (2.0)

Table 3. Classification accuracy (%) comparing different poolings. The largest gain came from AvgPool.

Dataset	Model	MaxPool	NNPool	AvgPool
Oxford-IIIT Pet	Val. Acc. w/ Noise	42.0 (1.1)	44.2 (2.8)	50.2 (1.9)
	Test Acc. w/ Noise	41.8 (0.9)	42.3 (3.2)	49.3 (1.8)
Caltech-101	Val. Acc. w/ Noise	59.5 (1.0)	58.3 (1.1)	62.7 (1.8)
	Test Acc. w/ Noise	57.2 (1.3)	56.7 (1.1)	60.8 (1.9)
FGVC-Aircraft	Val. Acc. w/ Noise	24.2 (3.5)	22.8 (1.9)	41.0 (2.9)
	Test Acc. w/ Noise	27.3 (3.6)	24.7 (1.8)	43.1 (3.1)
Caltech-UCSD Birds-200-2011	Val. Acc. w/ Noise	26.9 (1.8)	27.5 (3.0)	28.8 (1.8)
	Test Acc. w/ Noise	26.1 (1.7)	25.6 (2.6)	26.8 (1.2)
Stanford Cars	Val. Acc. w/ Noise	43.3 (3.4)	49.1 (1.9)	52.1 (1.8)
	Test Acc. w/ Noise	42.2 (2.8)	46.9 (1.3)	51.2 (2.1)

of CLIP pretrained weights due to different datasets or training schemes, but we present a different perspective on this issue.

We performed ablation studies to identify what determined the difference in robustness (Appendix F). We discovered that the core factor in different robustness arose from the preprocessing pipeline. Specifically, CLIP ViTs apply mean-std normalization to input images using certain per-channel mean and standard deviation (std) constants, which we refer to as the OPENAI constants (Appendix A), whereas supervised ViTs apply different per-channel mean-std constants, which are often called INCEPTION constants (Szegedy et al., 2016). In other words, the OPENAI mean-std constants led to vulnerability to Gaussian noise, whereas the INCEPTION mean-std constants did not show this vulnera-

bility.

Indeed, when we replaced the OPENAI mean-std constants with the INCEPTION constants, the CLIP ViTs achieved improved robustness (Table 5). The reverse also holds, and similar vulnerability was observed when adopting IMAGENET mean-std constants for ViTs. Full results on other datasets are available in Appendix D, where we observed these improvements across various pretrained weights with different training recipes.

6.2. Theoretical Analysis

We give an explanation for the empirical vulnerability of CLIP ViTs to additive Gaussian noise. The key point is that channel-wise normalization sets the pixel-space sensitivity scale: Smaller per-channel stds in the input normalization enlarge the worst-case response to perturbations even before the backbone acts.

Setup. Let $x \in [0, 1]^{C \times H \times W}$ be an image and η an additive perturbation. Let $\mu \in \mathbb{R}^C$ and $\sigma \in \mathbb{R}_{>0}^C$ be the per-channel means and stds, and define the normalization $N_{\mu, \sigma}(x) := (x - \mu)/\sigma$. Let $f : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^K$ denote the vision backbone operating on normalized inputs, which is globally ℓ_2 -Lipschitz with constant L_z on its domain.¹

¹This assumption holds when linear layers have bounded spectral norms and other modules are Lipschitz. ReLU: 1-Lipschitz (Gouk et al., 2021); GELU: ≈ 1.13 (Hendrycks & Gimpel, 2016); LayerNorm: Lipschitz with a constant set by γ and ε (Ba et al., 2016).

Table 4. ImageNet-1K results for ViT-B/16 224² with eight different pretrained weights. CLIP ViTs tended to yield worse ranks under noise.

Pretrained Model	Mean-Std	Top-1 $\rightarrow$ w/ Noise	Rank $\rightarrow$ w/ Noise	RankDiff
vit_base_patch16_224.augreg_in1k	INCEPTION	79.15 $\rightarrow$ 62.21	862 $\rightarrow$ 487	-375
vit_base_patch16_224.augreg2_in21k_ft_in1k	INCEPTION	85.10 $\rightarrow$ 74.50	203 $\rightarrow$ 96	-107
vit_base_patch16_224.orig_in21k_ft_in1k	INCEPTION	81.79 $\rightarrow$ 60.91	603 $\rightarrow$ 513	-90
vit_base_patch16_224.augreg_in21k_ft_in1k	INCEPTION	84.53 $\rightarrow$ 71.19	257 $\rightarrow$ 192	-65
vit_base_patch16_clip_224.openai_ft_in12k_in1k	OPENAI	85.94 $\rightarrow$ 70.81	135 $\rightarrow$ 209	+74
vit_base_patch16_clip_224.laion2b_ft_in12k_in1k	OPENAI	86.17 $\rightarrow$ 71.24	114 $\rightarrow$ 189	+75
vit_base_patch16_clip_224.laion2b_ft_in1k	OPENAI	85.47 $\rightarrow$ 67.88	168 $\rightarrow$ 311	+143
vit_base_patch16_clip_224.openai_ft_in1k	OPENAI	85.29 $\rightarrow$ 67.06	182 $\rightarrow$ 340	+158

Table 5. Classification accuracy (%) for fine-tuning ViTs on the Oxford-IIIT Pet.

Pretrained Model	Mean-Std	Val. Acc. w/ Noise	Test Acc. w/ Noise
vit_base_patch16_clip_224.openai_ft_in12k_in1k	OPENAI	94.5 (1.0) $\rightarrow$ 77.7 (3.4)	93.8 (1.0) $\rightarrow$ 76.3 (4.1)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	INCEPTION	95.5 (0.5) $\rightarrow$ 87.3 (2.1)	95.2 (0.6) $\rightarrow$ 87.2 (2.2)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	IMAGENET	94.2 (0.4) $\rightarrow$ 73.9 (2.3)	93.4 (0.5) $\rightarrow$ 72.7 (2.5)
vit_base_patch16_clip_224.datacomp_xl	OPENAI	93.6 (0.9) $\rightarrow$ 67.4 (6.0)	93.2 (0.9) $\rightarrow$ 67.3 (5.9)
vit_base_patch16_clip_224.datacomp_xl	INCEPTION	94.7 (0.5) $\rightarrow$ 78.5 (4.0)	93.6 (0.6) $\rightarrow$ 78.4 (3.8)
vit_base_patch16_clip_224.datacomp_xl	IMAGENET	92.8 (0.9) $\rightarrow$ 57.6 (7.4)	92.6 (0.5) $\rightarrow$ 58.1 (7.4)
vit_base_patch16_clip_224.dfn2b	OPENAI	95.0 (0.3) $\rightarrow$ 73.1 (1.5)	94.1 (0.5) $\rightarrow$ 73.3 (1.9)
vit_base_patch16_clip_224.dfn2b	INCEPTION	94.8 (0.8) $\rightarrow$ 78.6 (4.9)	93.6 (0.4) $\rightarrow$ 79.8 (5.0)
vit_base_patch16_clip_224.dfn2b	IMAGENET	95.1 (0.3) $\rightarrow$ 69.8 (2.7)	94.0 (0.4) $\rightarrow$ 68.8 (3.1)
vit_base_patch16_clip_224.metaclip_2pt5b	OPENAI	92.8 (0.7) $\rightarrow$ 64.8 (4.4)	92.0 (0.7) $\rightarrow$ 62.3 (3.9)
vit_base_patch16_clip_224.metaclip_2pt5b	INCEPTION	94.7 (0.4) $\rightarrow$ 78.5 (2.0)	93.9 (0.3) $\rightarrow$ 78.5 (1.8)
vit_base_patch16_clip_224.metaclip_2pt5b	IMAGENET	91.6 (0.3) $\rightarrow$ 54.5 (2.5)	90.8 (0.3) $\rightarrow$ 52.8 (1.6)
vit_base_patch16_clip_224.openai	OPENAI	92.5 (0.3) $\rightarrow$ 71.7 (1.0)	91.9 (0.6) $\rightarrow$ 70.2 (1.2)
vit_base_patch16_clip_224.openai	INCEPTION	94.0 (0.7) $\rightarrow$ 78.6 (4.6)	93.2 (0.9) $\rightarrow$ 77.3 (5.1)
vit_base_patch16_clip_224.openai	IMAGENET	91.2 (0.5) $\rightarrow$ 58.5 (4.0)	90.7 (0.8) $\rightarrow$ 58.4 (4.3)
vit_base_patch16_clip_224.laion2b	OPENAI	91.8 (1.2) $\rightarrow$ 56.1 (7.7)	90.5 (1.1) $\rightarrow$ 54.0 (6.6)
vit_base_patch16_clip_224.laion2b	INCEPTION	93.8 (0.6) $\rightarrow$ 76.4 (1.9)	92.8 (0.5) $\rightarrow$ 75.6 (1.8)
vit_base_patch16_clip_224.laion2b	IMAGENET	90.2 (0.8) $\rightarrow$ 52.3 (4.4)	89.5 (0.8) $\rightarrow$ 51.4 (4.1)
vit_base_patch16_224.augreg_in1k	OPENAI	95.5 (0.2) $\rightarrow$ 88.7 (0.3)	94.9 (0.2) $\rightarrow$ 88.2 (0.7)
vit_base_patch16_224.augreg_in1k	INCEPTION	95.5 (0.1) $\rightarrow$ 89.7 (0.5)	94.4 (0.3) $\rightarrow$ 89.2 (0.8)
vit_base_patch16_224.augreg_in1k	IMAGENET	95.5 (0.2) $\rightarrow$ 87.7 (0.5)	94.9 (0.2) $\rightarrow$ 87.9 (0.7)
vit_base_patch16_224.augreg_in21k	OPENAI	95.6 (0.3) $\rightarrow$ 91.4 (0.3)	95.2 (0.5) $\rightarrow$ 91.9 (0.6)
vit_base_patch16_224.augreg_in21k	INCEPTION	95.9 (0.2) $\rightarrow$ 92.3 (0.3)	95.6 (0.4) $\rightarrow$ 92.6 (0.4)
vit_base_patch16_224.augreg_in21k	IMAGENET	95.7 (0.5) $\rightarrow$ 91.6 (0.5)	95.6 (0.3) $\rightarrow$ 92.0 (0.5)
vit_base_patch16_224.mae	OPENAI	93.5 (0.3) $\rightarrow$ 70.8 (2.8)	93.4 (0.2) $\rightarrow$ 72.7 (2.3)
vit_base_patch16_224.mae	INCEPTION	93.7 (0.3) $\rightarrow$ 75.0 (2.1)	93.3 (0.2) $\rightarrow$ 75.2 (2.5)
vit_base_patch16_224.mae	IMAGENET	93.5 (0.3) $\rightarrow$ 72.0 (2.0)	92.7 (0.5) $\rightarrow$ 71.9 (2.2)

We study the end-to-end pipeline $F_{\mu,\sigma} := f \circ N_{\mu,\sigma}$ and its ℓ_2 -Lipschitz constant $\|F_{\mu,\sigma}\|_{\text{Lip}}$.

Theorem 6.1 (Pixel-space Lipschitz bound). *For any image x and perturbation η , we obtain*

$$\|F_{\mu,\sigma}(x + \eta) - F_{\mu,\sigma}(x)\|_2 \leq L_z \left\| \frac{\eta}{\sigma} \right\|_2 \leq \frac{L_z}{\sigma_{\min}} \|\eta\|_2,$$

where $\sigma_{\min} := \min_c \sigma_c$. In particular, the pixel-space Lipschitz constant satisfies $\|F_{\mu,\sigma}\|_{\text{Lip}} \leq L_z / \sigma_{\min}$.

Proof. Write $z = N_{\mu,\sigma}(x)$ and $\tilde{z} = N_{\mu,\sigma}(x + \eta) = z +$

η/σ . By Lipschitzness of f , we have $\|f(\tilde{z}) - f(z)\|_2 \leq L_z \|\eta/\sigma\|_2 \leq (L_z / \sigma_{\min}) \|\eta\|_2$. $\square$

Remark 6.2 (Practical reading of Theorem 6.1). For the standard choices

$$\begin{aligned} \sigma_{\text{INCEPTION}} &= (0.5, 0.5, 0.5), \\ \sigma_{\text{CLIP}} &= (0.26862954, 0.26130258, 0.27577711), \end{aligned}$$

the worst-case pixel-space sensitivity bound for CLIP is greater by a factor

$$\frac{L_z / \min(\sigma_{\text{CLIP}})}{L_z / \min(\sigma_{\text{INCEPTION}})} = \frac{0.5}{0.26130258} \approx 1.91,$$

relative to a supervised ViT using INCEPTION statistics. This $\sim 1.91\times$ looser bound amplifies the effect of input perturbations before the feature extractor.

7. Conclusion

Across `timm` models and controlled experiments, four design and pipeline factors consistently improved robustness against Gaussian noise: (1) larger stem kernel sizes, (2) resolution-dependent downsampling under pre-resize sensor noise, (3) average pooling instead of max pooling, and (4) supervised ViTs rather than CLIP ViTs. Practically, we recommend models with these properties such as `vit_base_patch32_224_augreg_in21k_ft_in1k` for ViT-B as an example. Our analysis integrates these findings: Theorem 4.1 proves that noise attenuation is quadratic with stem kernel size; Theorem 4.3 yields an analogous gain from anti-aliased downsampling in the input pipeline; Theorem 5.1 shows that average pooling is unbiased with error that falls as the window grows, whereas max pooling is positively biased and, for a uniform signal, its error grows logarithmically; and Theorem 6.1 explains CLIP sensitivity using pixel-space Lipschitz bounds scaling as $1/\sigma_{\min}$, which leads to a $\sim 1.91\times$ difference when comparing the OPENAI and INCEPTION constants. These insights provide actionable guidelines for practitioners to enhance the robustness of vision models against Gaussian noise in diverse applications.

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Impact Statement

We present work whose goal is to advance the field of machine learning. Our work has potential societal consequences, but we do not see any that require specific highlighting here.

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A. Experimental Setup

Here, we present the experimental details and full hyperparameters for the implementations.

Gaussian Noise We injected Gaussian noise into images using the `GaussNoise()` function from the `Albumentations` library (Buslaev et al., 2020). By default, we used the transform `A.GaussNoise(std_range=(0.1, 0.22), p=1.0)` with a scale factor with range (0.1, 0.22), which determines the fraction of the maximum value, *i.e.*, 255 for uint8 images or 1.0 for float images. For ImageNet-1K experiments, we used a scale factor with a range of (0.2, 0.44). The probability of applying Gaussian noise was set to 1. Note that Gaussian noise was applied only during evaluation, *i.e.*, during the test phase, not during the training phase.

ResNet Experiments We targeted multi-class classification tasks on the Oxford-IIIT Pet, Caltech-101, FGVC-Aircraft, Caltech-UCSD Birds-200-2011, and Stanford Cars datasets. The Oxford-IIIT Pet dataset contains 7K pet images from 37 classes; the Caltech-101 dataset includes 9K object images from 101 classes with a background category; the FGVC-Aircraft dataset includes 10K aircraft images from 102 classes; the Caltech-UCSD Birds-200-2011 dataset includes 12K bird images from 200 classes; and the Stanford Cars dataset includes 16K car images from 196 classes. These datasets are publicly available on their official websites. Each dataset was split into training, validation, and test sets with a ratio of 70:15:15. Unless specified otherwise, all experiments were conducted at a resolution of 224^2 using standard data augmentation, including random resized cropping to 256 pixels, random rotations within 15 degrees, color jitter with a factor of 0.4, random

horizontal flip with a probability of 0.5, center cropping with 224-pixel windows, and mean-std normalization based on ImageNet statistics.

For training, stochastic gradient descent with a momentum of 0.9, learning rate of 0.01, cosine annealing schedule with 200 iterations (Loshchilov & Hutter, 2017), weight decay of 10^{-2} , and mini-batch size of 128 were used. These hyperparameters were determined based on the accuracy of the validation set. One exception was made for experiments with larger resolutions ranging from 224^2 to 896^2 , where we used mini-batch size of 64 to adjust GPU memory, while other hyperparameters are the same. The model with the highest validation accuracy was obtained after 200 training epochs, and we reported accuracy on the validation and test sets. The ResNets were trained from scratch to solely focus on the architectural difference. The training was conducted on a single GPU machine. An average and standard deviation of five runs with different random seeds were reported for each result.

For ResNet, we used five types with the following architectures:

- Original ResNet: 7×7 stem with a width = 64 with single-layer, strided convolution in downsampling.
- ResNet-C: 3-layer 3×3 stem with a width = 32 (32, 32, 64), strided convolution in downsampling.
- ResNet-D: 3-layer 3×3 stem with a width = 32 (32, 32, 64), average pool in downsampling.
- ResNet-S: 3-layer 3×3 stem with a width = 64 (64, 64, 128), strided convolution in downsampling.
- ResNet-T: 3-layer 3×3 stem with a width = 32 (24, 48, 64), average pool in downsampling.

CLIP Experiments For the CLIP experiments, we used pretrained weights for both supervised ViTs and CLIP ViTs. When performing fine-tuning experiments, we used a learning rate of 0.001 and a weight decay of 2×10^{-4} , while keeping all other hyperparameters the same as in the above setup in ResNet.

ImageNet-1K Training The ImageNet-1K dataset contains 1.28M images for 1,000 classes. We referred to the hyperparameter recipe described in the official documentation and the recipe from DeiT (Touvron et al., 2021). For training, the AdamW optimizer (Loshchilov & Hutter, 2019) with learning rate 5×10^{-4} , epochs 400, warm-up learning rate 10^{-6} , cosine annealing schedule (Loshchilov & Hutter, 2017), weight decay 0.05, label smoothing (Szegedy et al., 2016) 0.1, RandAugment (Cubuk et al., 2020) of magnitude 9 and noise-std 0.5 with increased severity (rand-m9-mstd0.5-inc1), random erasing (Zhong et al., 2020) with probability 0.25, Cutmix (Yun et al., 2019) 1.0, stochastic depth (Huang et al., 2016) 0.1, mini-batch size 128 per GPU, Exponential Moving Average of model weights with decay factor 0.99996, and image resolution 224^2 were used. The training was performed on a $4 \times A100$ GPU machine, which required two to three days per training.

Mean-Std Constants Note that pretrained models may have been trained by any of the normalization constants; our choice of mean-std constants was applied on evaluation or fine-tuning of pretrained models. For training our own models, mean-std constants were applied during both the training and test phases. The exact values are as follows:

```
OPENAI_CLIP_MEAN = (0.48145466, 0.4578275, 0.40821073)
OPENAI_CLIP_STD = (0.26862954, 0.26130258, 0.27577711)
IMAGENET_INCEPTION_MEAN = (0.5, 0.5, 0.5)
IMAGENET_INCEPTION_STD = (0.5, 0.5, 0.5)
IMAGENET_DEFAULT_MEAN = (0.485, 0.456, 0.406)
IMAGENET_DEFAULT_STD = (0.229, 0.224, 0.225)
```

B. Rank Difference as a Robustness Proxy

Here, we denote the rank difference (RankDiff) at severity $\tau > 0$,

$$\text{RankDiff}_i(\tau) := \text{rank}_\tau(i) - \text{rank}_0(i),$$

where rank_τ orders models by accuracy at τ , so a more negative RankDiff_i indicates a robustness gain. In this section, we show that RankDiff is a principled, scale-free proxy because it aggregates pairwise rank flips caused by robustness slope differences.

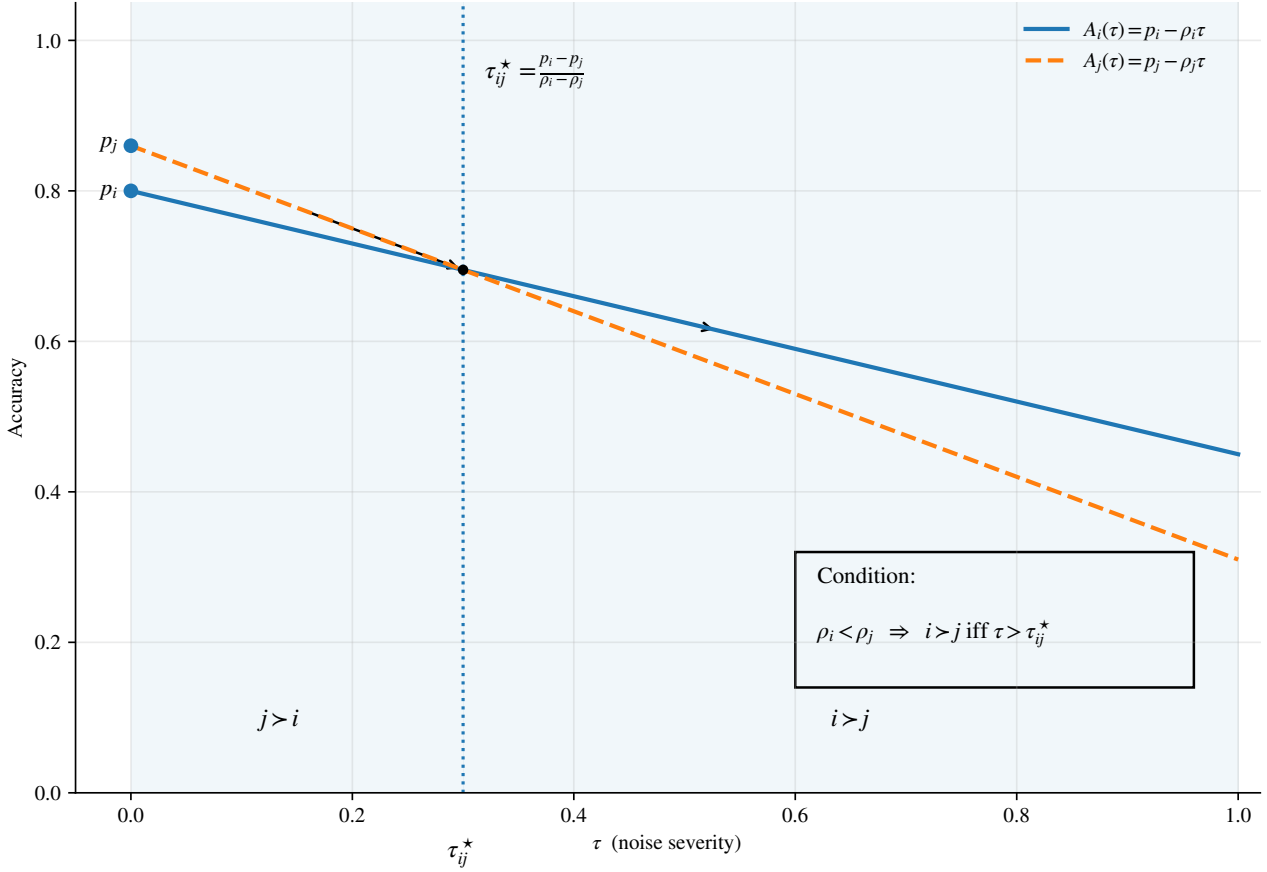


Figure 2. Illustration of a rank flip

Assumption (Local Linearity with Quadratic Remainder). For model $i \in \{1, \dots, M\}$, let $A_i(\tau)$ be its accuracy at noise severity $\tau \geq 0$ and $p_i := A_i(0)$. For some $\tau_0 > 0$,

$$A_i(\tau) = p_i - \rho_i \tau + r_i(\tau), \quad \rho_i \geq 0, \quad |r_i(\tau)| \leq L_i \tau^2 \quad (\tau \in [0, \tau_0]), \quad (5)$$

where ρ_i is the first-order robustness slope, and L_i bounds the curvature. The linear accuracy drop after applying a specific corruption has been verified in several studies (Recht et al., 2019; Hendrycks & Dietterich, 2019).

Pairwise Flip Rule. For any $i \neq j$,

$$A_i(\tau) - A_j(\tau) = (p_i - p_j) - (\rho_i - \rho_j)\tau + \varepsilon_{ij}(\tau), \quad |\varepsilon_{ij}(\tau)| \leq (L_i + L_j)\tau^2. \quad (6)$$

If $\rho_i \neq \rho_j$, the first-order flip threshold is

$$\tau_{ij}^* := \frac{p_i - p_j}{\rho_i - \rho_j}. \quad (7)$$

When $\tau_{ij}^* \in (0, \tau_0]$ and the margin condition

$$|(p_i - p_j) - (\rho_i - \rho_j)\tau| > (L_i + L_j)\tau^2 \quad (8)$$

holds at τ , the sign of $A_i(\tau) - A_j(\tau)$ is determined by the first-order term: Model i outranks j at τ if and only if $\tau > \tau_{ij}^*$ when $\rho_i < \rho_j$ (Figure 2).

RankDiff Counts Robustness-Driven Flips. Let $\mathcal{A}_i(\tau) := \{j \neq i : |(p_i - p_j) - (\rho_i - \rho_j)\tau| \leq (L_i + L_j)\tau^2\}$ be the set of ambiguous pairs at τ . Then, we have

$$|\text{RankDiff}_i(\tau) + \sum_{j \neq i} \text{sgn}(\rho_j - \rho_i) \mathbf{1}\{\rho_i \neq \rho_j, 0 < \tau_{ij}^* \leq \tau\}| \leq |\mathcal{A}_i(\tau)|. \quad (9)$$

In particular, if Eq. 8 holds for all $j \neq i$ at τ , equality holds in Eq. 9: $\text{RankDiff}_i(\tau)$ equals the net number of pairwise flips caused by having a smaller slope ρ_i .

Under the empirically observed near-linearity of accuracy-severity curves within the tested range, RankDiff is a scale-free robustness score: It ignores absolute calibration of accuracies and rewards models with smaller slopes ρ_i by counting the robustness-driven improvements in relative order.

Although we wrote that a negative RankDiff indicates better robustness, we are not saying that understanding the absolute value of RankDiff would capture robustness; to clarify our approach, we rather compare pairwise architectures to compute their corresponding $\Delta\text{RankDiff}$ to understand the relative difference in robustness.

C. List of Notations

Table 6. Kernel and resolution-related notations.

Symbol	Description
$x \in [0, 1]^{C \times H \times W}$	Input image with C channels, height H , width W .
$\eta \sim \mathcal{N}(0, \sigma^2 I)$	Additive i.i.d. Gaussian noise with per-pixel std σ .
I, I_n	Identity matrix of appropriate size; $I_n \in \mathbb{R}^{n \times n}$.
$*$	2D discrete convolution.
$\hat{u}$	DFT of u on the grid Ω .
Ω	DFT grid.
ε	Infrared cutoff $\varepsilon = 2\pi / \max\{H, W\}$.
$K_k \in \mathbb{R}^{k \times k}$	Stem kernel of side length k ; $\hat{K}_k$ denotes its DFT.
$\phi_k(r) = (1 + \beta kr)^{-1-\delta}$	Radial low-pass envelope upper-bounding $ \hat{K}_k(\omega) $.
β, δ	Positive envelope constants.
$\gamma(k) = \frac{\mathbb{E}\ K_k * \eta\ _2^2}{\sigma^2 HW}$	Per-pixel noise gain of the stem; equals $\ K_k\ _F^2$.
$s \geq 1$	Downsampling factor.
$g(s)$	Anti-alias filter size before downsampling; $c_1 s \leq g(s) \leq c_2 s$.
c_1, c_2	Absolute positive constants, independent of s .
$D_s = (\downarrow_s) \circ K_{g(s)}$	Anti-aliased downsampling: Filter then downsample by s .
$\downarrow_s$	Downsampling by a factor s along height and width.
$\gamma_{\downarrow}(s) = \frac{\mathbb{E}\ D_s \eta\ _2^2}{\sigma^2 HW/s^2}$	Per-output-pixel noise gain after downsampling.
$\mathbb{E}[\cdot], \text{Var}[\cdot]$	Expectation and variance.
C, C'	Absolute constants independent of k and s in the bounds.
$\ \cdot\ _2, \ \cdot\ _{\infty}, \ \cdot\ _F$	Euclidean, sup, and Frobenius norms.

Table 7. Pooling and CLIP-related notations.

Symbol	Description
$w, m = w^2$	Pooling window side length and number of elements.
$S = (S_1, \dots, S_m)$	Clean activations in one pooling window; $S_{(j)}$ denotes the j -th order statistic.
$X_{\text{avg}} = \frac{1}{m} \sum_{i=1}^m (S_i + \eta_i)$	Average-pooled noisy activation.
$X_{\text{max}} = \max_{1 \leq i \leq m} (S_i + \eta_i)$	Max-pooled noisy activation.
$S_{\text{avg}} = \frac{1}{m} \sum_i S_i, S_{\text{max}} = \max_i S_i$	Clean pooled activations.
$\delta_{\text{avg}} = X_{\text{avg}} - S_{\text{avg}}$	Avg-pool error; $\mathbb{E}[\delta_{\text{avg}}] = 0, \text{Var}[\delta_{\text{avg}}] = \sigma^2/m$.
$\delta_{\text{max}} = X_{\text{max}} - S_{\text{max}}$	Max-pool error.
$T_{\text{avg}}, T_{\text{max}}$	Pooling maps on a window for average and max.
$\ T\ _{\ell_2 \rightarrow \ell_2}$	Lipschitz constant in ℓ_2 ; $\ T_{\text{avg}}\ = m^{-1/2}, \ T_{\text{max}}\ \leq 1$.
$\Delta = S_{(1)} - S_{(2)}$	Gap between the largest and second-largest clean entries.
$Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$	Standard normals; $M_m = \max_i Z_i, A_m = \max_i Z_i $.
$\mu \in \mathbb{R}^C, \sigma \in \mathbb{R}_{>0}^C$	Per-channel mean and std for input normalization.
$N_{\mu, \sigma}(x) = (x - \mu)/\sigma$	Channel-wise normalization.
f	Vision backbone operating on normalized inputs.
$z = N_{\mu, \sigma}(x)$	Normalized input.
L_z	Global ℓ_2 -Lipschitz constant of f on its domain.
$F_{\mu, \sigma} = f \circ N_{\mu, \sigma}$	End-to-end map; $\ F_{\mu, \sigma}\ _{\text{Lip}} \leq L_z/\sigma_{\min}$.
$\sigma_{\min} = \min_c \sigma_c$	Smallest channel std in normalization.

Table 8. Rank difference-related notations.

Symbol	Description
$\tau \geq 0$	Noise severity level.
$A_i(\tau)$	Accuracy of model i at severity τ ; $p_i = A_i(0)$ denotes clean accuracy.
ρ_i	First-order accuracy slope with respect to severity.
L_i, τ_0	Curvature bound and validity radius for the local model.
$\text{rank}_{\tau}(i)$	Rank of model i by accuracy at severity τ .
$\text{RankDiff}_i(\tau) = \text{rank}_{\tau}(i) - \text{rank}_0(i)$	Rank change.
$\tau_{ij}^* = \frac{p_i - p_j}{\rho_i - \rho_j}$	First-order crossing severity of models i and j .
$\mathcal{A}_i(\tau)$	Set of j whose ordering with i is ambiguous at τ .
$\text{sgn}(\cdot), \mathbf{1}\{\cdot\}$	Sign and indicator functions.

D. Additional Experimental Results

Counterfactual Resolution Controls. To separate preprocessing from intrinsic backbone effects, we ran matched 224-vs-384 counterfactual controls on seven supervised ViT pairs across Flowers102, Oxford-IIIT Pet, and ImageNet-1K. We compared the default `pre_native` protocol, where noise is added before each model’s native resize, with two post-resize controls: `post_native`, where noise is added after each native resize, and `post_u224`, where both models share a 256-to-224 pipeline before the 384 model is upsampled to 384 and noise is added after the final resize. As shown in Table 9, moving noise after resizing reverses the apparent low-resolution advantage. Thus, the lower native resolution trend under the default protocol should be interpreted primarily as a downsampling effect, not as intrinsic robustness of lower-resolution backbones.

Table 9. Counterfactual 224-vs-384 resolution controls over seven matched supervised ViT pairs. Gaps are reported as 384 minus 224. Consistency is averaged over Flowers102, Oxford-IIIT Pet, and ImageNet-1K across three noise levels; noisy accuracy is measured on ImageNet-1K.

Metric	<code>pre_native</code>	<code>post_native</code>	<code>post_u224</code>
Mean consistency gap, # positive	−0.50 (34/63)	+4.18 (62/63)	+1.31 (47/63)
ImageNet noisy top-1 gap, # positive	−	+2.74 (21/21)	+2.33 (20/21)

Resolution Sweep and Boundary. We also swept the evaluation resolution on ImageNet-1K using the matched ViT-B/16 224 and 384 checkpoints, with resolutions in $\{160, 192, 224, 256, 288, 320, 352, 384\}$ and noise levels $\sigma \in \{0.02, 0.05, 0.10\}$. As shown in Table 10, lower resolution is not uniformly better: aggressive downsampling eventually loses task-relevant image information, so the beneficial regime is checkpoint- and image-dependent. In this sweep, the near-optimal bands were 224/256/288/320 for the 224 checkpoint and 352/384 for the 384 checkpoint.

A simple margin certificate makes this boundary explicit. If the classifier evaluated at a resized input has Lipschitz constant L_r and clean logit margin $m_r(x)$, then after anti-aliased downsampling by s , Theorem 4.3 and Markov’s inequality give a label-flip probability bounded up to constants by $L_r^2 C' \sigma^2 / (s^2 m_r(x)^2)$. Thus, downsampling improves robustness only while the s^{-2} noise reduction dominates the resolution-induced decrease in clean margin; because $m_r(x)$ and L_r depend on the image and checkpoint, the boundary is a regime condition rather than a universal pixel value.

Table 10. ImageNet-1K resolution sweep for matched ViT-B/16 checkpoints. Best resolutions are reported for noise levels $\sigma = 0.02/0.05/0.10$. The near-optimal band summarizes resolutions within 0.5 percentage points of the mean optimum across the sweep.

Checkpoint	Best consistency res.	Best noisy top-1 res.	Near-optimal band
224 checkpoint	192/224/192	320/256/288	224/256/288/320
384 checkpoint	256/320/320	352/384/352	352/384

Figure 3 shows the accuracy on the validation set for the controlled experiments on kernel size and resolution.

We also report additional results for ResNet-50-D (Table 11) and ResNet-101-D (Table 12) for different choices of pooling.

Table 13 summarizes the results for ResNet-AA, which adopts anti-aliasing average pooling architecture (Zhang, 2019). Specifically, ResNet-AA adopts average pooling in all downsampling layers as well as replacing the max pooling in the stem with average pooling. ResNet-AA was marginally more robust than the ResNet with average pooling only in the stem, but not as significant as the difference with the original ResNet. The result indicates that the core difference in robustness was caused by the use of average pooling in the stem.

Table 14 summarizes ImageNet-1K results for other ViT configurations, including different patch sizes, resolutions, and training recipes.

Table 15, Table 16, Table 17, and Table 18 summarize full results for fine-tuning ViTs on other datasets. When we replaced the OPENAI mean-std constants with the INCEPTION constants, the CLIP ViTs achieved improved robustness.

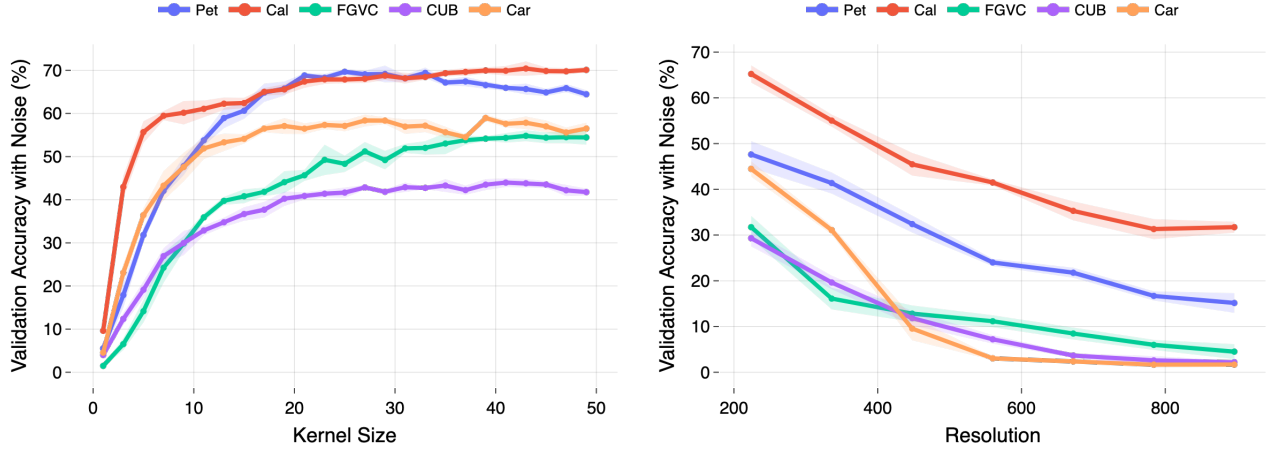


Figure 3. The results on the validation set

Table 11. Classification accuracy comparing different poolings, using ResNet-50-D

Dataset	Model	MaxPool	NNPool	AvgPool
Oxford-IIIT Pet	Val. Acc.	87.7 (0.6)	87.6 (0.4)	86.7 (0.5)
	Test Acc.	85.3 (0.8)	84.7 (0.6)	84.8 (0.9)
	Val. Acc. w/ Noise	48.3 (2.2)	46.3 (1.9)	54.0 (3.7)
	Test Acc. w/ Noise	47.8 (1.3)	45.2 (2.3)	53.6 (2.7)
Caltech-101	Val. Acc.	81.3 (0.7)	82.4 (1.1)	81.7 (0.5)
	Test Acc.	80.5 (0.3)	80.7 (0.4)	81.6 (0.7)
	Val. Acc. w/ Noise	61.1 (1.5)	60.3 (1.4)	62.7 (1.4)
	Test Acc. w/ Noise	59.8 (1.7)	58.3 (1.3)	61.6 (1.3)
FGVC-Aircraft	Val. Acc.	68.1 (0.2)	67.7 (0.8)	69.0 (0.7)
	Test Acc.	68.8 (1.1)	68.3 (1.5)	69.6 (0.3)
	Val. Acc. w/ Noise	27.7 (1.6)	24.8 (1.8)	42.9 (1.7)
	Test Acc. w/ Noise	31.5 (2.1)	26.9 (0.8)	44.8 (1.1)
Caltech-UCSD Birds-200-2011	Val. Acc.	69.8 (0.7)	69.8 (0.4)	69.3 (1.1)
	Test Acc.	67.3 (0.4)	66.4 (0.6)	65.9 (0.4)
	Val. Acc. w/ Noise	26.8 (0.6)	28.7 (1.7)	31.8 (1.6)
	Test Acc. w/ Noise	26.0 (0.7)	27.4 (1.2)	31.1 (2.1)
Stanford Cars	Val. Acc.	86.5 (0.5)	85.7 (0.5)	84.9 (0.2)
	Test Acc.	84.8 (0.2)	83.6 (0.3)	83.2 (0.3)
	Val. Acc. w/ Noise	56.0 (0.5)	53.6 (1.6)	56.8 (2.2)
	Test Acc. w/ Noise	54.8 (1.5)	51.6 (1.5)	55.3 (2.0)

Table 12. Classification accuracy comparing different poolings, using ResNet-101-D

Dataset	Model	MaxPool	NNPool	AvgPool
Oxford-IIIT Pet	Val. Acc.	87.0 (0.5)	86.5 (0.8)	86.2 (0.3)
	Test Acc.	84.8 (0.6)	84.4 (0.7)	84.3 (0.6)
	Val. Acc. w/ Noise	52.3 (1.9)	51.0 (1.5)	56.4 (2.1)
	Test Acc. w/ Noise	51.0 (1.3)	49.2 (1.4)	56.3 (2.4)
Caltech-101	Val. Acc.	82.0 (0.9)	82.9 (0.6)	82.9 (0.5)
	Test Acc.	80.6 (0.4)	80.7 (0.9)	81.2 (0.4)
	Val. Acc. w/ Noise	63.4 (1.9)	63.7 (1.3)	64.8 (1.1)
	Test Acc. w/ Noise	62.1 (1.5)	61.6 (1.7)	63.7 (1.4)
FGVC-Aircraft	Val. Acc.	69.5 (0.3)	67.7 (0.6)	69.4 (0.8)
	Test Acc.	71.0 (1.0)	67.1 (0.4)	69.6 (0.7)
	Val. Acc. w/ Noise	36.9 (4.0)	28.5 (2.7)	48.4 (1.5)
	Test Acc. w/ Noise	39.1 (3.5)	30.5 (2.6)	49.5 (1.8)
Caltech-UCSD Birds-200-2011	Val. Acc.	70.5 (0.5)	70.0 (0.7)	68.9 (0.6)
	Test Acc.	67.4 (0.7)	66.8 (0.4)	66.0 (0.7)
	Val. Acc. w/ Noise	29.7 (1.7)	29.3 (2.0)	33.4 (1.8)
	Test Acc. w/ Noise	29.0 (1.7)	29.2 (2.6)	32.2 (1.5)
Stanford Cars	Val. Acc.	84.5 (0.4)	83.9 (0.4)	83.7 (0.5)
	Test Acc.	83.3 (0.2)	81.9 (0.8)	82.1 (0.6)
	Val. Acc. w/ Noise	57.5 (1.2)	55.2 (0.9)	58.2 (1.2)
	Test Acc. w/ Noise	56.0 (0.9)	54.5 (0.7)	56.4 (1.2)

Table 13. Results on ResNet-AA

Dataset	Model	ResNet-AA-50	ResNet-AA-50-D	ResNet-AA-101-D
Oxford-IIIT Pet	Val. Acc.	84.8 (0.8)	86.9 (0.4)	86.2 (0.3)
	Test Acc.	83.1 (0.8)	84.7 (0.8)	84.3 (0.2)
	Val. Acc. w/ Noise	50.1 (2.7)	55.6 (2.0)	58.1 (2.7)
	Test Acc. w/ Noise	49.6 (3.2)	53.9 (1.4)	58.9 (2.9)
Caltech-101	Val. Acc.	80.2 (0.4)	81.7 (0.7)	83.0 (0.3)
	Test Acc.	79.5 (0.6)	80.6 (0.5)	80.9 (0.5)
	Val. Acc. w/ Noise	61.2 (1.6)	61.7 (2.2)	65.0 (1.5)
	Test Acc. w/ Noise	60.1 (1.5)	60.8 (2.8)	63.3 (1.3)
FGVC-Aircraft	Val. Acc.	67.3 (0.5)	69.8 (1.0)	69.1 (0.6)
	Test Acc.	67.1 (0.9)	70.7 (1.2)	70.0 (0.9)
	Val. Acc. w/ Noise	40.4 (3.6)	45.5 (2.5)	49.0 (2.9)
	Test Acc. w/ Noise	42.3 (3.9)	48.3 (2.2)	49.5 (2.5)
Caltech-UCSD Birds-200-2011	Val. Acc.	65.3 (0.6)	68.9 (0.8)	69.4 (0.6)
	Test Acc.	62.3 (1.1)	66.1 (0.6)	66.1 (0.4)
	Val. Acc. w/ Noise	28.6 (0.8)	32.5 (1.0)	31.7 (2.8)
	Test Acc. w/ Noise	27.5 (1.3)	31.4 (1.9)	31.0 (2.4)
Stanford Cars	Val. Acc.	79.9 (0.6)	85.9 (0.3)	83.5 (0.6)
	Test Acc.	78.9 (0.6)	83.9 (0.4)	81.6 (0.8)
	Val. Acc. w/ Noise	51.8 (1.6)	60.3 (2.8)	57.2 (3.2)
	Test Acc. w/ Noise	50.3 (1.0)	58.9 (2.0)	56.0 (3.2)

Table 14. ImageNet-1K results for other ViT configurations

Pretrained Model	Mean-Std	Top-1 → w/ Noise	Rank → w/ Noise	RankDiff
vit_base_patch16_384.augreg.in1k	INCEPTION	81.10 → 60.23	676 → 524	-152
vit_base_patch16_384.augreg.in21k.ft.in1k	INCEPTION	85.99 → 70.89	129 → 208	+79
vit_base_patch16_clip_384.laion2b.ft.in12k.in1k	OPENAI	87.21 → 70.38	55 → 227	+172
vit_base_patch16_clip_384.openai.ft.in1k	OPENAI	86.20 → 68.55	110 → 285	+175
vit_base_patch16_clip_384.openai.ft.in12k.in1k	OPENAI	87.03 → 69.11	61 → 269	+208
vit_base_patch16_clip_384.laion2b.ft.in1k	OPENAI	86.62 → 66.63	83 → 348	+265
vit_base_patch32_224.augreg.in1k	INCEPTION	74.90 → 58.44	1075 → 569	-506
vit_base_patch32_224.sam.in1k	INCEPTION	73.69 → 51.33	1101 → 748	-353
vit_base_patch32_224.augreg.in21k.ft.in1k	INCEPTION	80.71 → 65.31	719 → 392	-327
vit_base_patch32_clip_224.openai.ft.in1k	OPENAI	81.93 → 63.94	591 → 428	-163
vit_base_patch32_clip_224.laion2b.ft.in1k	OPENAI	82.58 → 63.09	504 → 450	-54
vit_base_patch32_clip_224.laion2b.ft.in12k.in1k	OPENAI	83.30 → 65.57	419 → 386	-33
vit_base_patch32_384.augreg.in1k	INCEPTION	78.75 → 59.65	893 → 539	-354
vit_base_patch32_384.augreg.in21k.ft.in1k	INCEPTION	83.35 → 63.72	412 → 437	+25
vit_base_patch32_clip_384.openai.ft.in12k.in1k	OPENAI	85.21 → 68.40	191 → 293	+102
vit_base_patch32_clip_384.laion2b.ft.in12k.in1k	OPENAI	85.37 → 65.58	180 → 383	+203

Table 15. Classification accuracy (%) for fine-tuning ViTs on the Caltech-101.

Pretrained Model	Mean-Std	Val. Acc. w/ Noise	Test Acc. w/ Noise
vit_base_patch16_clip_224_openai_ft_in12k_in1k	OPENAI	93.1 (0.6) → 84.1 (1.1)	92.0 (0.8) → 81.8 (1.4)
vit_base_patch16_clip_224_openai_ft_in12k_in1k	INCEPTION	95.7 (0.6) → 90.4 (0.8)	94.5 (0.7) → 89.5 (1.2)
vit_base_patch16_clip_224_openai_ft_in12k_in1k	IMAGENET	91.6 (1.2) → 80.5 (2.4)	90.5 (0.8) → 78.5 (2.4)
vit_base_patch16_clip_224_datacomp_xl	OPENAI	95.3 (0.8) → 86.4 (2.3)	94.6 (0.6) → 84.8 (2.1)
vit_base_patch16_clip_224_datacomp_xl	INCEPTION	96.2 (0.6) → 91.0 (1.3)	95.7 (0.9) → 89.7 (1.4)
vit_base_patch16_clip_224_datacomp_xl	IMAGENET	94.7 (0.7) → 82.5 (1.9)	93.8 (1.0) → 80.8 (2.6)
vit_base_patch16_clip_224_dfn2b	OPENAI	90.2 (11.2) → 80.1 (15.0)	88.9 (12.3) → 78.8 (14.8)
vit_base_patch16_clip_224_dfn2b	INCEPTION	96.5 (0.6) → 91.7 (1.2)	95.9 (0.5) → 91.0 (1.8)
vit_base_patch16_clip_224_dfn2b	IMAGENET	93.7 (3.9) → 79.9 (10.5)	92.4 (4.6) → 78.2 (10.7)
vit_base_patch16_clip_224_metaclip_2pt5b	OPENAI	94.9 (0.7) → 81.5 (2.0)	94.2 (0.7) → 79.5 (2.0)
vit_base_patch16_clip_224_metaclip_2pt5b	INCEPTION	96.0 (0.5) → 89.5 (2.1)	95.0 (0.8) → 87.8 (2.8)
vit_base_patch16_clip_224_metaclip_2pt5b	IMAGENET	93.6 (1.0) → 76.3 (3.2)	92.3 (1.2) → 74.6 (2.9)
vit_base_patch16_clip_224_openai	OPENAI	92.8 (0.2) → 78.9 (3.1)	91.7 (1.1) → 76.9 (3.6)
vit_base_patch16_clip_224_openai	INCEPTION	95.4 (0.3) → 87.8 (0.9)	95.4 (0.6) → 86.9 (0.9)
vit_base_patch16_clip_224_openai	IMAGENET	92.3 (0.4) → 80.3 (1.8)	91.8 (0.7) → 77.7 (1.9)
vit_base_patch16_clip_224_laion2b	OPENAI	92.3 (0.9) → 77.7 (2.4)	91.2 (0.6) → 75.6 (1.6)
vit_base_patch16_clip_224_laion2b	INCEPTION	95.3 (0.6) → 87.3 (0.3)	94.3 (0.6) → 85.8 (0.5)
vit_base_patch16_clip_224_laion2b	IMAGENET	90.1 (0.8) → 71.5 (2.4)	89.2 (0.5) → 67.6 (2.4)
vit_base_patch16_224_augreg_in1k	OPENAI	94.4 (0.3) → 84.8 (0.9)	94.1 (0.3) → 85.7 (0.4)
vit_base_patch16_224_augreg_in1k	INCEPTION	94.1 (0.3) → 86.0 (0.5)	93.8 (0.2) → 86.7 (0.8)
vit_base_patch16_224_augreg_in1k	IMAGENET	94.3 (0.6) → 84.7 (0.6)	94.0 (0.3) → 85.7 (0.7)
vit_base_patch16_224_augreg_in21k	OPENAI	97.0 (0.4) → 95.1 (0.5)	96.3 (0.4) → 94.5 (0.7)
vit_base_patch16_224_augreg_in21k	INCEPTION	97.1 (0.3) → 95.8 (0.5)	96.6 (0.2) → 95.4 (0.3)
vit_base_patch16_224_augreg_in21k	IMAGENET	97.2 (0.2) → 95.1 (0.2)	96.6 (0.5) → 94.6 (0.5)
vit_base_patch16_224_mae	OPENAI	92.0 (0.5) → 76.3 (0.7)	91.6 (0.8) → 75.7 (1.2)
vit_base_patch16_224_mae	INCEPTION	91.6 (0.6) → 80.8 (1.2)	91.7 (0.4) → 79.4 (0.7)
vit_base_patch16_224_mae	IMAGENET	91.7 (0.5) → 75.4 (0.6)	91.6 (0.4) → 74.5 (1.2)

Table 16. Classification accuracy (%) for fine-tuning ViTs on the FGVC-Aircraft.

Pretrained Model	Mean-Std	Val. Acc. w/ Noise	Test Acc. w/ Noise
vit_base_patch16_clip_224.openai_ft_in12k_in1k	OPENAI	62.6 (1.7) → 46.6 (1.9)	61.7 (1.2) → 47.4 (1.7)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	INCEPTION	60.4 (23.6) → 50.8 (20.4)	59.5 (23.7) → 50.9 (21.1)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	IMAGENET	59.5 (1.4) → 44.0 (1.6)	58.2 (1.2) → 45.2 (1.2)
vit_base_patch16_clip_224.datacomp_xl	OPENAI	73.7 (4.9) → 50.7 (7.7)	72.2 (4.0) → 52.7 (7.1)
vit_base_patch16_clip_224.datacomp_xl	INCEPTION	80.8 (1.9) → 66.3 (3.7)	79.4 (2.0) → 66.3 (3.5)
vit_base_patch16_clip_224.datacomp_xl	IMAGENET	65.9 (4.1) → 40.0 (6.1)	65.0 (3.4) → 41.5 (5.2)
vit_base_patch16_clip_224.dfn2b	OPENAI	75.4 (4.9) → 55.7 (7.0)	75.2 (5.2) → 57.3 (8.0)
vit_base_patch16_clip_224.dfn2b	INCEPTION	82.0 (4.1) → 70.0 (8.1)	81.7 (4.3) → 70.7 (7.6)
vit_base_patch16_clip_224.dfn2b	IMAGENET	72.9 (6.6) → 51.3 (9.4)	71.3 (7.1) → 52.5 (9.8)
vit_base_patch16_clip_224.metaclip_2pt5b	OPENAI	68.0 (2.7) → 48.4 (4.1)	67.3 (2.5) → 49.7 (3.0)
vit_base_patch16_clip_224.metaclip_2pt5b	INCEPTION	80.5 (1.6) → 68.4 (3.2)	79.2 (2.2) → 69.5 (3.4)
vit_base_patch16_clip_224.metaclip_2pt5b	IMAGENET	64.5 (1.3) → 40.9 (2.4)	64.2 (1.4) → 43.3 (2.3)
vit_base_patch16_clip_224.openai	OPENAI	63.7 (4.7) → 47.4 (5.6)	61.9 (4.3) → 49.1 (4.4)
vit_base_patch16_clip_224.openai	INCEPTION	74.6 (3.5) → 65.4 (4.5)	73.4 (3.8) → 66.0 (5.4)
vit_base_patch16_clip_224.openai	IMAGENET	60.3 (1.6) → 42.6 (2.7)	59.4 (1.4) → 43.4 (2.6)
vit_base_patch16_clip_224.laion2b	OPENAI	59.9 (1.9) → 37.7 (2.4)	58.4 (1.7) → 38.5 (1.9)
vit_base_patch16_clip_224.laion2b	INCEPTION	69.2 (4.4) → 54.3 (5.5)	68.9 (5.4) → 55.0 (6.0)
vit_base_patch16_clip_224.laion2b	IMAGENET	58.3 (1.8) → 36.0 (2.3)	56.9 (1.3) → 37.3 (2.4)
vit_base_patch16_224.augreg_in1k	OPENAI	67.8 (0.8) → 50.7 (1.9)	67.0 (1.2) → 51.2 (1.7)
vit_base_patch16_224.augreg_in1k	INCEPTION	67.0 (0.5) → 52.4 (1.4)	67.2 (0.9) → 53.6 (1.0)
vit_base_patch16_224.augreg_in1k	IMAGENET	67.4 (0.4) → 50.1 (1.4)	67.3 (0.8) → 51.0 (2.5)
vit_base_patch16_224.augreg_in21k	OPENAI	78.2 (0.3) → 69.9 (0.5)	77.2 (0.6) → 69.4 (1.1)
vit_base_patch16_224.augreg_in21k	INCEPTION	78.6 (0.6) → 71.6 (0.4)	77.3 (0.4) → 71.0 (0.4)
vit_base_patch16_224.augreg_in21k	IMAGENET	77.8 (0.6) → 68.9 (0.9)	77.1 (1.0) → 68.5 (1.2)
vit_base_patch16_224.mae	OPENAI	69.3 (0.7) → 39.9 (4.2)	68.8 (1.5) → 40.3 (4.4)
vit_base_patch16_224.mae	INCEPTION	69.1 (0.7) → 43.5 (2.8)	69.1 (0.9) → 44.0 (2.3)
vit_base_patch16_224.mae	IMAGENET	69.1 (0.6) → 40.0 (2.1)	69.4 (1.2) → 41.8 (1.2)

Table 17. Classification accuracy (%) for fine-tuning ViTs on the Caltech-UCSD Birds-200-2011.

Pretrained Model	Mean-Std	Val. Acc. w/ Noise	Test Acc. w/ Noise
vit_base_patch16_clip_224.openai_ft_in12k_in1k	OPENAI	84.0 (0.9) → 64.0 (1.7)	81.3 (1.0) → 61.1 (1.1)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	INCEPTION	85.3 (1.6) → 69.3 (1.7)	82.7 (1.3) → 67.0 (2.5)
vit_base_patch16_clip_224.openai_ft_in12k_in1k	IMAGENET	82.6 (0.8) → 59.8 (1.3)	79.7 (1.6) → 56.7 (1.8)
vit_base_patch16_clip_224.datacomp_x1	OPENAI	83.4 (1.1) → 53.6 (2.3)	81.4 (1.0) → 50.7 (2.6)
vit_base_patch16_clip_224.datacomp_x1	INCEPTION	84.7 (0.7) → 59.7 (4.5)	82.8 (0.8) → 57.3 (3.8)
vit_base_patch16_clip_224.datacomp_x1	IMAGENET	83.6 (0.9) → 52.2 (2.8)	81.5 (1.1) → 49.3 (2.6)
vit_base_patch16_clip_224.dfn2b	OPENAI	84.8 (1.2) → 58.8 (2.6)	83.0 (1.3) → 56.4 (2.3)
vit_base_patch16_clip_224.dfn2b	INCEPTION	87.3 (1.6) → 69.6 (4.7)	86.0 (2.0) → 67.3 (5.2)
vit_base_patch16_clip_224.dfn2b	IMAGENET	81.6 (2.7) → 50.0 (2.3)	79.7 (2.8) → 48.1 (2.9)
vit_base_patch16_clip_224.metaclip_2pt5b	OPENAI	83.3 (0.5) → 49.5 (3.5)	81.1 (0.9) → 47.9 (3.2)
vit_base_patch16_clip_224.metaclip_2pt5b	INCEPTION	85.8 (0.9) → 62.1 (2.0)	83.4 (0.6) → 60.1 (1.8)
vit_base_patch16_clip_224.metaclip_2pt5b	IMAGENET	81.3 (2.5) → 45.3 (4.5)	78.7 (2.7) → 43.6 (4.2)
vit_base_patch16_clip_224.openai	OPENAI	83.4 (0.5) → 60.1 (2.4)	81.8 (0.8) → 57.7 (2.8)
vit_base_patch16_clip_224.openai	INCEPTION	85.5 (0.8) → 66.7 (3.4)	83.3 (1.3) → 65.1 (3.7)
vit_base_patch16_clip_224.openai	IMAGENET	75.3 (14.1) → 50.1 (13.8)	72.7 (13.9) → 47.5 (12.6)
vit_base_patch16_clip_224.laion2b	OPENAI	81.4 (1.4) → 52.1 (2.2)	78.5 (2.5) → 50.0 (2.0)
vit_base_patch16_clip_224.laion2b	INCEPTION	84.6 (0.6) → 62.0 (2.1)	82.2 (0.4) → 59.9 (2.1)
vit_base_patch16_clip_224.laion2b	IMAGENET	81.0 (0.4) → 50.1 (0.7)	78.7 (0.5) → 48.3 (1.1)
vit_base_patch16_clip_224.augreg_in1k	OPENAI	83.4 (0.4) → 67.6 (0.7)	81.7 (0.8) → 65.8 (0.7)
vit_base_patch16_224.augreg_in1k	INCEPTION	83.9 (0.5) → 69.3 (0.7)	81.8 (0.3) → 67.8 (0.5)
vit_base_patch16_224.augreg_in1k	IMAGENET	83.7 (0.4) → 67.5 (1.2)	81.8 (0.2) → 65.9 (0.7)
vit_base_patch16_224.augreg_in21k	OPENAI	89.6 (0.2) → 84.0 (0.5)	88.9 (0.5) → 83.4 (0.4)
vit_base_patch16_224.augreg_in21k	INCEPTION	89.6 (0.2) → 84.9 (0.7)	88.7 (0.4) → 83.7 (0.6)
vit_base_patch16_224.augreg_in21k	IMAGENET	89.5 (0.2) → 83.9 (0.1)	88.9 (0.3) → 83.4 (0.7)
vit_base_patch16_224.mae	OPENAI	76.7 (0.5) → 39.3 (4.4)	74.1 (0.6) → 36.7 (3.9)
vit_base_patch16_224.mae	INCEPTION	74.0 (0.3) → 41.1 (4.2)	72.5 (1.1) → 38.9 (4.4)
vit_base_patch16_224.mae	IMAGENET	76.4 (0.9) → 38.0 (1.5)	74.4 (0.5) → 35.6 (1.7)

E. Correlation Analysis

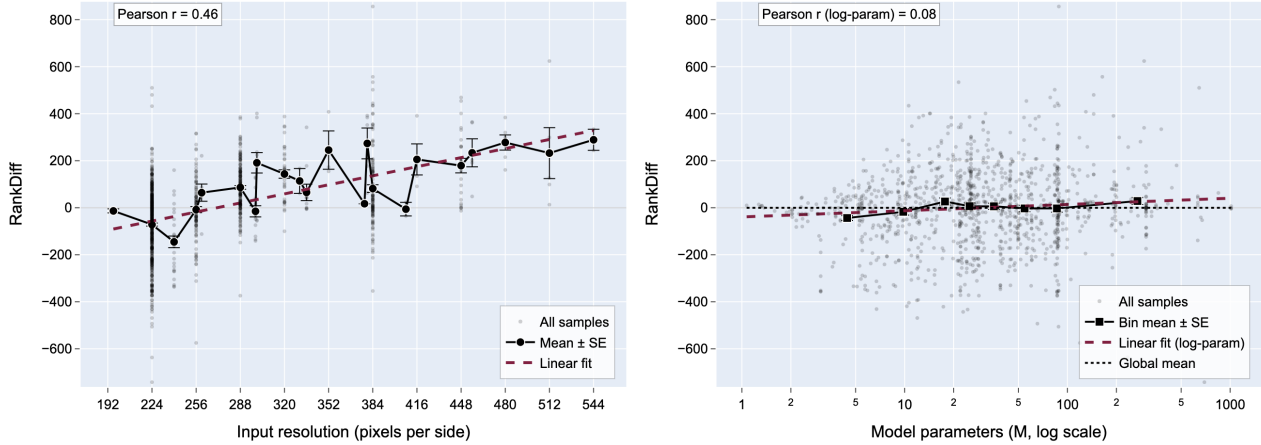


Figure 4. For all 1,174 timm models, we plotted the rank difference with respect to resolution (left) and the number of parameters (right). A significant level of correlation was found for resolution.

The experiments in the main text targeted ViTs and ResNets with a controlled setup for kernel size and resolution. Here, we further found that these observations hold for other vision models. Figure 4 summarizes how the rank difference is affected by resolution and the number of parameters across all 1,174 timm vision models, which also include other models beyond ViTs and ResNets. Firstly, we observed that the number of parameters showed no relationship with the rank difference, which implies that choosing a larger model does not lead to improved robustness against Gaussian noise. By contrast, we

Table 18. Classification accuracy (%) for fine-tuning ViTs on the Stanford-Cars.

Pretrained Model	Mean-Std	Val. Acc. w/ Noise	Test Acc. w/ Noise
vit_base_patch16_clip_224_openai_ft_in12k_in1k	OPENAI	83.8 (0.1) → 71.0 (1.4)	83.0 (0.7) → 69.7 (0.7)
vit_base_patch16_clip_224_openai_ft_in12k_in1k	INCEPTION	87.3 (1.2) → 77.7 (2.1)	86.2 (1.3) → 76.3 (2.2)
vit_base_patch16_clip_224_openai_ft_in12k_in1k	IMAGENET	81.1 (1.6) → 63.8 (2.2)	80.7 (1.9) → 64.6 (2.2)
vit_base_patch16_clip_224_datacomp_xl	OPENAI	90.1 (0.7) → 76.1 (1.7)	89.2 (0.6) → 75.3 (1.5)
vit_base_patch16_clip_224_datacomp_xl	INCEPTION	91.3 (0.2) → 80.9 (0.8)	90.4 (0.6) → 79.4 (1.0)
vit_base_patch16_clip_224_datacomp_xl	IMAGENET	89.8 (1.4) → 75.4 (3.8)	89.1 (1.4) → 74.3 (3.8)
vit_base_patch16_clip_224_dfn2b	OPENAI	91.1 (0.5) → 78.9 (2.5)	90.2 (0.5) → 77.8 (2.2)
vit_base_patch16_clip_224_dfn2b	INCEPTION	94.2 (1.1) → 88.7 (2.2)	93.2 (1.0) → 87.6 (2.8)
vit_base_patch16_clip_224_dfn2b	IMAGENET	91.1 (1.8) → 78.8 (5.0)	90.7 (1.4) → 77.6 (5.4)
vit_base_patch16_clip_224_metaclip_2pt5b	OPENAI	87.7 (0.7) → 67.7 (1.7)	86.9 (0.7) → 66.4 (1.7)
vit_base_patch16_clip_224_metaclip_2pt5b	INCEPTION	91.1 (0.3) → 78.5 (1.3)	90.2 (0.4) → 77.3 (1.6)
vit_base_patch16_clip_224_metaclip_2pt5b	IMAGENET	87.1 (1.3) → 64.7 (2.0)	86.1 (1.7) → 63.2 (2.3)
vit_base_patch16_clip_224_openai	OPENAI	85.6 (3.3) → 73.5 (4.1)	85.3 (3.1) → 72.4 (3.9)
vit_base_patch16_clip_224_openai	INCEPTION	89.8 (0.4) → 81.0 (1.1)	89.5 (0.4) → 80.2 (0.7)
vit_base_patch16_clip_224_openai	IMAGENET	85.2 (1.6) → 70.1 (3.0)	84.2 (1.3) → 69.0 (3.0)
vit_base_patch16_clip_224_laion2b	OPENAI	84.8 (2.4) → 65.6 (4.2)	84.1 (2.3) → 65.3 (3.7)
vit_base_patch16_clip_224_laion2b	INCEPTION	89.9 (0.8) → 78.4 (2.3)	88.8 (0.9) → 77.0 (2.2)
vit_base_patch16_clip_224_laion2b	IMAGENET	79.9 (4.7) → 54.5 (6.6)	79.5 (5.1) → 54.9 (7.6)
vit_base_patch16_224_augreg_in1k	OPENAI	82.8 (0.5) → 67.4 (1.0)	81.6 (0.4) → 66.3 (0.9)
vit_base_patch16_224_augreg_in1k	INCEPTION	83.2 (0.6) → 69.2 (1.1)	81.6 (0.5) → 67.5 (1.3)
vit_base_patch16_224_augreg_in1k	IMAGENET	83.0 (0.3) → 66.2 (1.4)	81.5 (0.2) → 65.1 (1.6)
vit_base_patch16_224_augreg_in21k	OPENAI	89.7 (0.2) → 82.6 (0.5)	88.5 (0.3) → 81.4 (0.5)
vit_base_patch16_224_augreg_in21k	INCEPTION	89.9 (0.2) → 84.2 (0.4)	88.3 (0.3) → 83.3 (0.7)
vit_base_patch16_224_augreg_in21k	IMAGENET	89.9 (0.5) → 81.9 (0.5)	88.6 (0.6) → 81.1 (0.3)
vit_base_patch16_224_mae	OPENAI	80.4 (0.5) → 61.1 (1.5)	78.0 (0.6) → 58.5 (0.9)
vit_base_patch16_224_mae	INCEPTION	80.3 (0.3) → 61.7 (1.0)	77.6 (0.5) → 59.3 (0.8)
vit_base_patch16_224_mae	IMAGENET	80.6 (0.4) → 58.1 (2.2)	78.3 (0.3) → 56.7 (2.4)

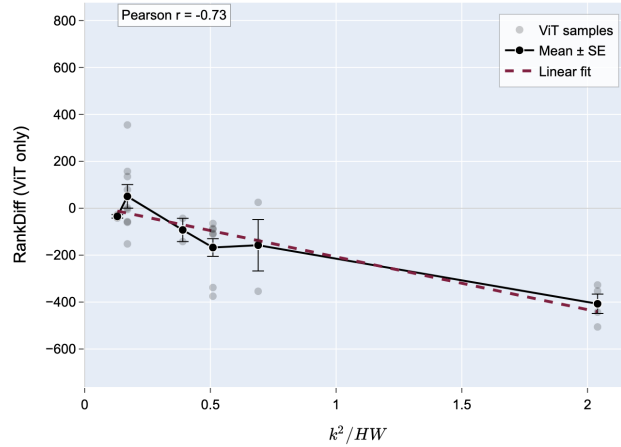


Figure 5. For ViTs, the rank difference exhibited a significant level of correlation with the ratio of patch to image.

observed that the resolution, as well as the kernel size, had a significant level of correlation with the rank difference. Note that this correlation arises even though there are plenty of other factors that affect robustness, such as different training recipes. Overall, smaller native resolutions were associated with smaller rank differences under the pre-resize sensor-noise protocol, consistent with the downsampling interpretation in the main text.

Extending the findings of the main text, we can also say that the ratio of patch to image has a significant correlation with robustness against Gaussian noise. For ViTs, this ratio becomes $100 \cdot k^2 / HW$ (%). We investigated its relationship with

the rank difference (Figure 5), targeting the ViTs listed in Table 1. Again, although these ViTs were trained with different recipes, the overall tendency showed a significant correlation: a higher ratio of patch to image led to a smaller rank difference, indicating improved robustness under this protocol.

F. Are There Other Factors That Cause Vulnerabilities of CLIP?

We investigated other factors that might possibly address the vulnerability of CLIP. However, the vulnerability of CLIP could not be fully addressed by other factors examined below.

How About Swapping Pretrained Weights with Supervised ViT? Answer: No. Differences in training datasets and losses would lead to different pretrained weights for CLIP ViTs. Assuming that certain dataset or loss properties, or equivalently certain properties of the pretrained weights of CLIP ViTs, lead to vulnerabilities, we performed controlled experiments to swap parts of them with those of supervised ViTs. Specifically, we swapped pretrained weights of each block in `vit_base_patch16_clip_224.openai` with those of `vit_base_patch16_224.augreg2_in21k_ft_in1k` to see which module weights determine the robustness against Gaussian noise (Table 19). Although swapping pretrained weights partially addressed the vulnerability of CLIP ViTs in certain cases near the last block such as targeting block12, the improvements were not as significant as the approach of replacing mean-std constants. Furthermore, the improvement depended on the specific weight choice in the target block; block12.mlp.fc2.weight improved robustness, whereas block12.norm1.weight did not. When we swapped multiple weights such as block12.{mlp.fc2, mlp.fc1, norm2}, the performance rather degraded, which indicates that improvement is not guaranteed.

Table 19. Results of swapping pretrained weights in CLIP ViT. The accuracy with Gaussian noise partially improved.

Swap	Val. Acc. → w/ Noise	Test Acc. → w/ Noise
stem	53.1 (1.2) → 36.3 (0.9)	51.3 (1.5) → 35.4 (1.2)
block1	81.3 (19.1) → 48.6 (14.8)	80.1 (19.9) → 46.2 (14.1)
block2	41.9 (3.4) → 20.1 (1.6)	40.7 (3.9) → 18.8 (1.6)
block3	68.6 (13.6) → 29.9 (4.5)	67.9 (13.7) → 28.7 (4.4)
block4	80.2 (6.8) → 29.6 (7.5)	79.8 (6.4) → 28.6 (7.9)
block5	77.0 (4.0) → 30.0 (4.0)	77.3 (3.2) → 29.0 (3.4)
block6	84.7 (0.8) → 39.9 (1.9)	84.0 (0.5) → 37.8 (2.2)
block7	87.8 (0.4) → 45.0 (0.8)	86.5 (0.6) → 44.9 (1.2)
block8	90.5 (0.4) → 49.9 (2.7)	88.4 (0.6) → 48.4 (1.6)
block9	90.7 (0.2) → 56.5 (3.9)	90.1 (0.4) → 54.8 (2.2)
block10	91.5 (0.4) → 62.3 (3.3)	91.0 (0.4) → 60.4 (3.5)
block11	91.4 (0.4) → 59.6 (5.1)	90.7 (0.9) → 58.1 (6.0)
block12	91.5 (0.5) → 62.3 (4.9)	91.4 (0.6) → 60.8 (4.4)
head	82.8 (7.6) → 48.4 (7.5)	82.3 (6.8) → 48.2 (6.6)
Baseline (IMAGENET)	91.2 (0.5) → 58.5 (4.0)	90.7 (0.8) → 58.4 (4.3)
Ours (INCEPTION)	92.5 (0.3) → 71.7 (1.0)	91.9 (0.6) → 70.2 (1.2)

How About Architectural Differences Such as norm_pre? Answer: No. Although the architecture is almost the same for CLIP ViT and supervised ViTs, one difference is that CLIP ViTs insert additional LayerNorm in the patch embedding before the transformer blocks start, which we refer to as norm_pre. Assuming that the use of norm_pre causes vulnerability, we performed controlled experiments training ViTs with and without norm_pre (Table 21). Nevertheless, the ViT with norm_pre, which corresponds to the identical architecture of CLIP ViTs, rather exhibited improved performance against Gaussian noise, which indicates that norm_pre does not lead to the vulnerability observed in CLIP ViTs.

Table 20. Results of swapping specific weights in block12. Swapping multiple weights did not ensure improved robustness.

Swap	Val. Acc. $\rightarrow$ w/ Noise	Test Acc. $\rightarrow$ w/ Noise
block12.norm1.weight	91.0 (1.3) $\rightarrow$ 56.8 (8.2)	90.1 (1.3) $\rightarrow$ 56.6 (8.6)
block12.norm1.bias	91.5 (0.9) $\rightarrow$ 59.5 (5.7)	90.7 (1.4) $\rightarrow$ 58.1 (6.1)
block12.attn.qkv.weight	91.0 (0.5) $\rightarrow$ 60.5 (1.3)	90.3 (0.8) $\rightarrow$ 59.0 (1.4)
block12.attn.qkv.bias	91.0 (0.9) $\rightarrow$ 58.7 (3.6)	90.0 (0.9) $\rightarrow$ 58.4 (3.7)
block12.attn.proj.weight	92.1 (0.6) $\rightarrow$ 59.8 (5.6)	91.3 (0.9) $\rightarrow$ 59.8 (5.8)
block12.attn.proj.bias	90.9 (1.0) $\rightarrow$ 58.3 (5.4)	90.2 (1.2) $\rightarrow$ 57.8 (5.0)
block12.norm2.weight	91.8 (0.7) $\rightarrow$ 62.7 (4.3)	90.7 (0.9) $\rightarrow$ 61.1 (3.7)
block12.norm2.bias	91.4 (0.9) $\rightarrow$ 60.8 (3.4)	90.8 (0.7) $\rightarrow$ 59.7 (3.4)
block12.mlp.fc1.weight	91.4 (1.0) $\rightarrow$ 61.0 (7.6)	91.0 (1.4) $\rightarrow$ 60.3 (7.0)
block12.mlp.fc1.bias	91.2 (1.3) $\rightarrow$ 58.3 (3.4)	90.4 (1.5) $\rightarrow$ 57.4 (3.7)
block12.mlp.fc2.weight	91.3 (0.4) $\rightarrow$ 65.2 (2.2)	90.5 (0.3) $\rightarrow$ 63.8 (2.1)
block12.mlp.fc2.bias	91.4 (0.7) $\rightarrow$ 58.8 (5.0)	90.7 (0.7) $\rightarrow$ 58.2 (5.4)
block12.mlp.fc2	90.8 (0.6) $\rightarrow$ 58.7 (3.2)	90.0 (0.4) $\rightarrow$ 57.7 (3.1)
block12.mlp.fc2 & mlp.fc1	91.9 (1.1) $\rightarrow$ 64.2 (4.3)	91.6 (0.8) $\rightarrow$ 63.8 (5.0)
block12.mlp.fc2 & mlp.fc1 & norm2	91.0 (0.5) $\rightarrow$ 55.5 (4.3)	90.0 (1.4) $\rightarrow$ 54.0 (5.2)

Table 21. Results on different ViT architectures with and without norm_pre. The use of norm_pre did not bring vulnerability.

Architecture	Top-1 $\rightarrow$ w/ Noise	Top-5 $\rightarrow$ w/ Noise
w/o norm_pre	77.76 $\rightarrow$ 47.15	93.84 $\rightarrow$ 68.96
w/ norm_pre	78.84 $\rightarrow$ 54.22	94.14 $\rightarrow$ 76.13

G. On Gaussian Noise

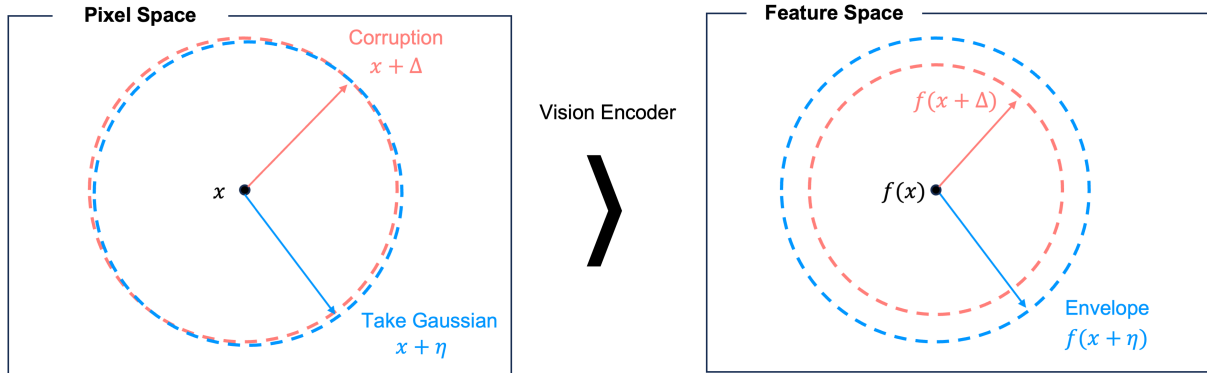


Figure 6. Illustration of Theorem 3.1. Features with Gaussian noise serve as an upper bound for corruption that has the same scale in pixel space.

There are several reasons why analyzing robustness against Gaussian noise is both useful and representative of common corruptions.

Gaussian Surrogate via Moment Matching. The local linearization Eq. 1 implies that the feature perturbation δ_f is well-approximated to first order in the input perturbation Δ . Let $\mu_\Delta := \mathbb{E}[\Delta]$ and $\Sigma_\Delta := \text{Cov}(\Delta)$, where the expectation is

taken over the randomness of the corruption. Plugging Eq. 1 into δ_f and taking expectations gives

$$\begin{aligned}\mathbb{E}[\delta_f] &= J_f(x)\mu_\Delta + O(\mathbb{E}\|\Delta\|_2^2), \\ \text{Cov}[\delta_f] &= J_f(x)\Sigma_\Delta J_f(x)^\top + O(\mathbb{E}\|\Delta\|_2^3).\end{aligned}$$

Thus, up to higher-order terms, any corruption whose pixel perturbation has mean μ_Δ and covariance Σ_Δ induces the same first two feature-space moments as the Gaussian feature perturbation $J_f(x)\eta$ generated by $\eta \sim \mathcal{N}(\mu_\Delta, \Sigma_\Delta)$. This applies both to zero-mean corruptions, such as noise and blur, and to mean-shifting ones, such as brightness enhancement, after decomposing Δ into its mean and zero-mean residual. In this sense, Gaussian noise serves as a convenient surrogate once we restrict attention to the low-order moments of the feature perturbation.

Gaussian Probes for Locally Linear Corruptions. We now show that Gaussian probes are, in fact, the worst-case within a broad variance-bounded family. Let a corruption $\mathcal{C}$ with randomness ξ produce $x' = \mathcal{C}(x, \xi)$ with $\Delta_{\mathcal{C}} := x' - x$, and assume that, for small severities, it admits a factorization

$$\Delta_{\mathcal{C}} = B_{\mathcal{C}}(x)\zeta, \quad \mathbb{E}[\zeta] = 0, \quad \text{Cov}(\zeta) \preceq I_d,$$

for some linear operator $B_{\mathcal{C}}(x)$ that depends smoothly on x and a random vector ζ . The covariance bound simply constrains the overall severity of the corruption. This model covers many image corruptions: Gaussian blur and motion blur correspond to convolutional $B_{\mathcal{C}}(x)$; brightness, contrast, and fog are locally affine rescalings; and JPEG compression or elastic distortions can be approximated as linear maps plus higher-order residuals at low severity.

Under the local linearization Eq. 1, we have $f(x + \Delta_{\mathcal{C}}) - f(x) \approx J_f(x)B_{\mathcal{C}}(x)\zeta$, and hence

$$\mathbb{E}[\|f(x + \Delta_{\mathcal{C}}) - f(x)\|_2^2] \approx \mathbb{E}[\|J_f(x)B_{\mathcal{C}}(x)\zeta\|_2^2] = \text{tr}(J_f(x)\Sigma_{\mathcal{C}}(x)J_f(x)^\top),$$

with $\Sigma_{\mathcal{C}}(x) := B_{\mathcal{C}}(x)\text{Cov}(\zeta)B_{\mathcal{C}}(x)^\top \preceq B_{\mathcal{C}}(x)B_{\mathcal{C}}(x)^\top$. Replacing ζ by $\eta \sim \mathcal{N}(0, I_d)$ yields

$$\mathbb{E}[\|f(x + B_{\mathcal{C}}(x)\eta) - f(x)\|_2^2] = \|J_f(x)B_{\mathcal{C}}(x)\|_F^2,$$

which saturates the same variance-bounded envelope: Any other zero-mean ζ with $\text{Cov}(\zeta) \preceq I_d$ can only decrease this expectation. Thus, once a corruption is reduced to a linear shape $B_{\mathcal{C}}(x)$, additive Gaussian noise with matching $B_{\mathcal{C}}(x)$ provides a worst-case, direction-agnostic stress test on f . Our architectural conclusions, such as kernel size, resolution, pooling, and normalization constants, depend only on how they scale this Jacobian-based quantity, so they transfer directly from Gaussian probes to a broad range of common corruptions that admit such local linear models.

Empirical Simulation Here, we performed an empirical simulation to investigate the validity of Theorem 3.1. Using common image corruptions, including blur, weather, and digital corruptions used in Hendrycks & Dietterich (2019), we first calibrated each corruption and Gaussian noise to have an equal maximum eigenvalue in pixel space and then compared the variance in feature space when passing through the same linear stem. For the stem, we considered five setups with different kernel sizes of 3, 5, and 7; high and low resolutions; and average pooling. Theoretically, Gaussian noise achieves an upper bound on this variance, and our simulations support this expectation: the ratio of corruption to Gaussian in feature-space variance saturates around 1 across all corruptions tested here (Figure 7). These results clearly demonstrate that the analysis of Gaussian noise captures the worst-case robustness against these common image corruptions.

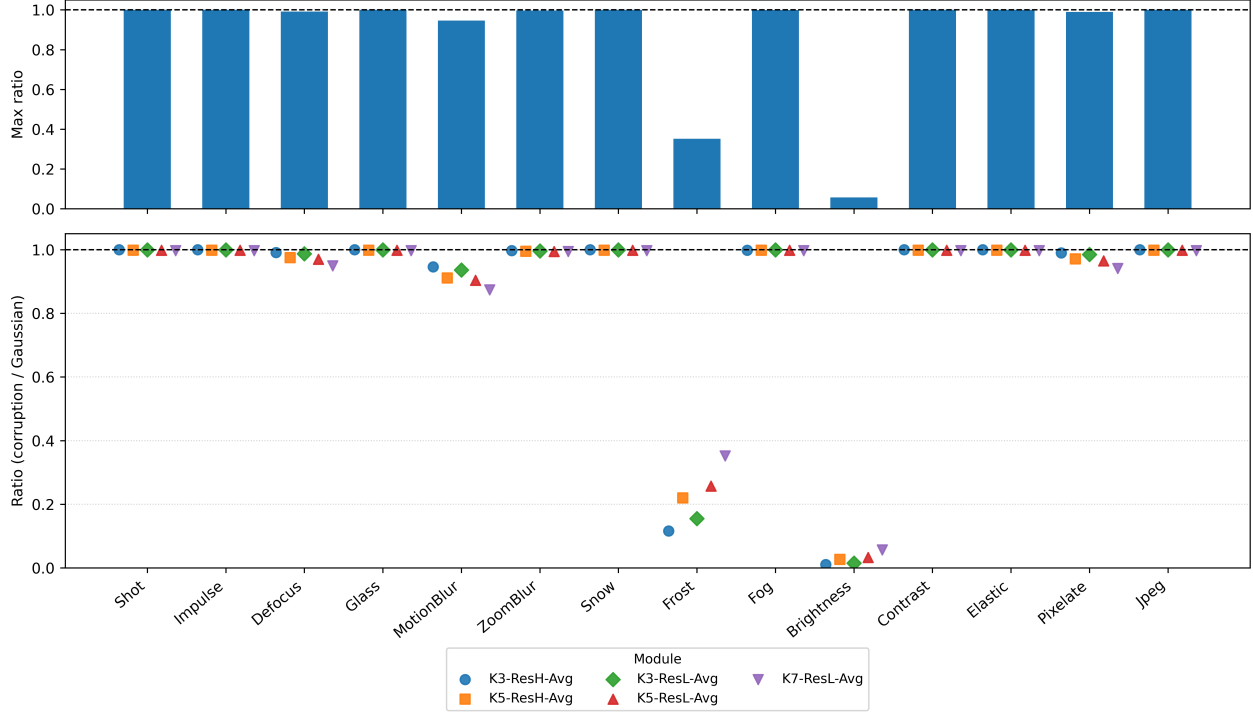


Figure 7. Empirical simulation of Theorem 3.1, comparing the variance in feature space for common image corruptions and Gaussian noise. All results show the ratio to be close to or less than one, which confirms that Gaussian noise serves as an upper bound.

H. Extension to Other Noise Models

We select Gaussian noise due to its approximation of aggregate perturbations by the central limit theorem and its prevalence in real-world imaging, such as sensor readout and thermal noise. Here, we explain how our main findings—noise attenuation by larger stem kernels and resolution-dependent downsampling (Theorems 4.1, 4.3), the pooling comparison (Theorem 5.1), and the normalization effect (Theorem 6.1)—extend beyond Gaussian noise.

Setup. We continue to use k for a filter side length. For pooling windows, we use w for side length and $m = w^2$ for the number of elements.

Poisson Noise. Let S be the filter support with $|S| = m = k^2$. Let $h = \{h_t\}_{t \in S}$ denote the linear filter coefficients on S , and $Y_t \sim \text{Poisson}(x_t)$ independent. For a locally constant intensity on the filter support, where $x_t \approx \bar{x}$ in a smooth patch, we have

$$\text{Var}\left[\sum_t h_t Y_t\right] = \sum_t h_t^2 \text{Var}[Y_t] = \sum_t h_t^2 x_t = \bar{x} \sum_t h_t^2 + \sum_t h_t^2 (x_t - \bar{x}) \approx \bar{x} \|h\|_2^2, \quad (10)$$

so the per-output-pixel variance inherits the k^{-2} and s^{-2} scalings up to the local factor $\bar{x}$. Applying the Anscombe transform $A(y) = 2\sqrt{y + 3/8}$ approximately stabilizes the Poisson variance to ≈ 1 , after which Gaussian-based methods are applicable (Anscombe, 1948).

Salt-and-Pepper Noise. Under the symmetric model where each pixel is replaced by either 0 or 1 with probability q and a locally constant patch with mean $\bar{x}$, we have

$$\mathbb{E}[\text{avg error}] = q(1/2 - \bar{x}), \quad \text{Var}[\text{avg error}] = O(1/m).$$

Max pooling tends to amplify these impulses. As a robust alternative, median pooling recovers the clean value in constant patches when contamination is lower than 50% and is 1-Lipschitz with respect to ℓ_∞ ; trimmed means are another option.

Normalization and Lipschitz Sensitivity. The pixel-space Lipschitz bound in Theorem 6.1 does not depend on the specific noise type, so smaller per-channel normalization stds increase the worst-case sensitivity equally for Gaussian and non-Gaussian perturbations.

I. Proof of Theorem 3.1

In this section, we prove the Gaussian envelope result stated in Theorem 3.1. Throughout, we fix x and write $J := J_f(x)$ for the Jacobian of f at x , and all expectations are taken with respect to the perturbation.

Proof. By the local linearization in Eq. 1, for any perturbation Δ , we have

$$f(x + \Delta) - f(x) = J\Delta + r(x, \Delta), \quad \|r(x, \Delta)\|_2 \leq \frac{L(x)}{2} \|\Delta\|_2^2, \quad (11)$$

for some local curvature bound $L(x) > 0$. For brevity, define $r := r(x, \Delta)$. Then

$$\|\delta_f\|_2^2 = \|J\Delta + r\|_2^2 = \|J\Delta\|_2^2 + 2\langle J\Delta, r \rangle + \|r\|_2^2. \quad (12)$$

Leading Term. Taking expectations and using $\mathbb{E}[\Delta] = 0$ with covariance $\Sigma_\Delta := \mathbb{E}[\Delta\Delta^\top]$, we obtain

$$\mathbb{E}[\|J\Delta\|_2^2] = \mathbb{E}[\Delta^\top J^\top J \Delta] = \text{tr}(J^\top J \mathbb{E}[\Delta\Delta^\top]) = \text{tr}(J^\top J \Sigma_\Delta) = \text{tr}(J \Sigma_\Delta J^\top). \quad (13)$$

Now use the spectral constraint $\Sigma_\Delta \preceq \sigma^2 I_d$. Let $B := J^\top J \succeq 0$ and write the eigen-decomposition $\Sigma_\Delta = Q\Lambda Q^\top$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $0 \leq \lambda_i \leq \sigma^2$. Then

$$\begin{aligned} \text{tr}(J \Sigma_\Delta J^\top) &= \text{tr}(B \Sigma_\Delta) = \text{tr}(B Q \Lambda Q^\top) = \text{tr}(\Lambda Q^\top B Q) \\ &= \sum_{i=1}^d \lambda_i (Q^\top B Q)_{ii} \leq (\max_i \lambda_i) \sum_{i=1}^d (Q^\top B Q)_{ii} \\ &\leq \sigma^2 \text{tr}(Q^\top B Q) = \sigma^2 \text{tr}(B) = \sigma^2 \|J\|_F^2. \end{aligned} \quad (14)$$

Combining Eq. 13 and Eq. 14 yields

$$\text{tr}(J \Sigma_\Delta J^\top) \leq \sigma^2 \|J\|_F^2. \quad (15)$$

Remainder Terms. Next, we control the second and third terms in Eq. 12 using the remainder bound in Eq. 11. First, by Cauchy–Schwarz,

$$|\langle J\Delta, r \rangle| \leq \|J\Delta\|_2 \|r\|_2 \leq \|J\|_F \|\Delta\|_2 \cdot \frac{L(x)}{2} \|\Delta\|_2^2 = \frac{L(x)}{2} \|J\|_F \|\Delta\|_2^3, \quad (16)$$

so

$$|\mathbb{E}[\langle J\Delta, r \rangle]| \leq \frac{L(x)}{2} \|J\|_F \mathbb{E}[\|\Delta\|_2^3]. \quad (17)$$

Similarly, from $\|r\|_2^2 \leq \frac{L(x)^2}{4} \|\Delta\|_2^4$, we have $\mathbb{E}[\|r\|_2^2] \leq \frac{L(x)^2}{4} \mathbb{E}[\|\Delta\|_2^4]$.

To interpret the big- O term, it is natural to consider a family of small perturbations $\Delta = \varepsilon \xi$ with a fixed random vector ξ and $\varepsilon > 0$ a scale parameter controlling the perturbation magnitude. Then $\mathbb{E}\|\Delta\|_2^3 = \varepsilon^3 \mathbb{E}\|\xi\|_2^3$ and $\mathbb{E}\|\Delta\|_2^4 = \varepsilon^4 \mathbb{E}\|\xi\|_2^4$, so we have

$$2|\mathbb{E}[\langle J\Delta, r \rangle]| + \mathbb{E}[\|r\|_2^2] = O(\varepsilon^3) = O(\mathbb{E}\|\Delta\|_2^3), \quad (18)$$

with a constant depending only on J , $L(x)$, and the law of ξ . We summarize this as $O(\mathbb{E}\|\Delta\|_2^3)$ in the statement of the theorem.

Putting Everything Together. Taking expectations in Eq. 12 and combining Eq. 13, Eq. 15, and Eq. 18, we obtain

$$\begin{aligned} \mathbb{E}[\|f(x + \Delta) - f(x)\|_2^2] &= \text{tr}(J \Sigma_\Delta J^\top) + O(\mathbb{E}\|\Delta\|_2^3) \\ &\leq \sigma^2 \|J\|_F^2 + O(\mathbb{E}\|\Delta\|_2^3), \end{aligned} \quad (19)$$

which proves the first claim.

Gaussian Case and Saturation. Now let $\eta \sim \mathcal{N}(0, \sigma^2 I_d)$. Then $\Sigma_\eta = \sigma^2 I_d$, and the leading term becomes

$$\text{tr}(J \Sigma_\eta J^\top) = \text{tr}(J(\sigma^2 I_d) J^\top) = \sigma^2 \text{tr}(J J^\top) = \sigma^2 \|J\|_F^2. \quad (20)$$

The same remainder analysis as above, applied with $\Delta = \eta$, yields

$$\mathbb{E}[\|f(x + \eta) - f(x)\|_2^2] = \sigma^2 \|J\|_F^2 + O(\mathbb{E}\|\eta\|_2^3). \quad (21)$$

Thus, among all zero-mean perturbations with covariance $\Sigma_\Delta \preceq \sigma^2 I_d$, Gaussian noise $\eta \sim \mathcal{N}(0, \sigma^2 I_d)$ saturates the upper bound on the leading Jacobian-based contribution to the expected feature-space mean-squared error. $\square$

J. Proofs for Theorems 4.1 and 4.3

Here, we provide proofs of the quadratic noise-decay results in Section 4.2.

J.1. Conventions and Assumptions

DFT Convention and Parseval. For $u \in \mathbb{R}^{H \times W}$ with discrete Fourier transform (DFT) $\hat{u}$ on the frequency grid Ω , we use the Parseval identity

$$\frac{1}{HW} \sum_{\omega \in \Omega} |\hat{u}(\omega)|^2 = \sum_{p \in \{1, \dots, H\} \times \{1, \dots, W\}} |u(p)|^2. \quad (22)$$

We write $\varepsilon := 2\pi / \max\{H, W\}$ for the infrared cutoff.

Filter Family. For $k \geq 3$, let $K_k \in \mathbb{R}^{k \times k}$ denote the linear, shift-invariant stem kernel with DFT $\hat{K}_k$. We assume only the following low-pass envelope; the same assumption applies to $K_{g(s)}$ when used as the anti-aliasing filter at scale $g(s)$:

- **(A_{roll})** (Radial low-pass envelope at scale $1/k$) There exist $\beta, \delta > 0$ such that, for all frequencies ω ,

$$|\hat{K}_k(\omega)| \leq \phi_k(\|\omega\|), \quad \phi_k(r) := (1 + \beta kr)^{-1-\delta},$$

where ϕ_k is nonincreasing in r .

This assumption provides a monotone radial upper envelope sufficient for establishing our upper bounds: When estimating $\frac{1}{HW} \sum_{\omega} |\hat{K}_k(\omega)|^2$, we first dominate $|\hat{K}_k|^2$ by ϕ_k^2 and then apply the sum-integral comparison in Eq. 26.

Noise Model and Gains. Let $\eta \sim \mathcal{N}(0, \sigma^2 I)$ be spatially white Gaussian noise. The per-pixel noise gain of the stem kernel is

$$\gamma(k) := \frac{\mathbb{E}[\|K_k * \eta\|_2^2]}{\sigma^2 HW} \stackrel{\text{Eq. 22}}{=} \frac{1}{HW} \sum_{\omega} |\hat{K}_k(\omega)|^2 = \|K_k\|_F^2. \quad (23)$$

For anti-aliased downsampling with a factor $s \geq 1$, we define

$$D_s := (\downarrow_s) \circ K_{g(s)}, \quad c_1 s \leq g(s) \leq c_2 s, \quad (24)$$

and its per-output-pixel noise gain

$$\gamma_{\downarrow}(s) := \frac{\mathbb{E}[\|D_s \eta\|_2^2]}{\sigma^2 HW / s^2}. \quad (25)$$

Radial Sum-Integral Comparison. Let Ω be the $H \times W$ DFT grid with spacing ε , and let $g : [\varepsilon, \pi] \rightarrow \mathbb{R}_{\geq 0}$ be radially nonincreasing. We partition Ω into annuli $\mathcal{A}_j := \{\omega : j\varepsilon \leq \|\omega\| < (j+1)\varepsilon\}$. Because each grid point occupies an area $\asymp \varepsilon^2$ and the annulus area is $2\pi r\varepsilon$ up to boundary effects, there exist absolute lattice constants $c_1, c_2 > 0$ —independent of H, W, k, s —with

$$c_1 HW (2\pi j\varepsilon) \leq |\mathcal{A}_j| \leq c_2 HW (2\pi (j+1)\varepsilon).$$

By monotonicity, $g((j+1)\varepsilon)|\mathcal{A}_j| \leq \sum_{\omega \in \mathcal{A}_j} g(\|\omega\|) \leq g(j\varepsilon)|\mathcal{A}_j|$. Summing over j and dividing by HW turns the lattice sum into upper and lower Riemann sums for $r \mapsto 2\pi r g(r)$ with mesh ε , yielding absolute constants $A_1, A_2 > 0$ such that

$$A_1 \int_{\varepsilon}^{\pi} r g(r) dr \leq \frac{1}{HW} \sum_{\omega \in \Omega} g(\|\omega\|) \leq A_2 \int_{\varepsilon}^{\pi} r g(r) dr. \quad (26)$$

As $\varepsilon \rightarrow 0$, both bounds converge to the same limit; for finite grids, A_1, A_2 absorb edge discrepancies and remain independent of the kernel scale k or downsampling factor s .

J.2. Proof of Theorem 4.1 (Quadratic Decay in Stem Kernel Size)

Proof. By Eq. 23, Eq. 26, and (A_{roll}) , we have

$$\gamma(k) \lesssim \int_{\varepsilon}^{\pi} r |\widehat{K}_k(r)|^2 dr \leq \int_{\varepsilon}^{\pi} r (1 + \beta k r)^{-2-2\delta} dr.$$

Let $u = 1 + \beta k r$. Then $r = (u - 1)/(\beta k)$ and $dr = du/(\beta k)$, so

$$\int_{\varepsilon}^{\pi} r (1 + \beta k r)^{-2-2\delta} dr = \frac{1}{\beta^2 k^2} \int_{1+\beta k \varepsilon}^{1+\beta k \pi} \frac{u-1}{u^{2+2\delta}} du \leq \frac{1}{\beta^2 k^2} \int_1^{\infty} \frac{u-1}{u^{2+2\delta}} du = \frac{C}{k^2},$$

for a finite constant $C = C(\beta, \delta)$. Hence $\gamma(k) \leq C'/k^2$ for some C' independent of k . $\square$

J.3. Proof of Theorem 4.3 (Quadratic Decay under Anti-Aliased Downsampling)

We first state the following identity for white noise.

Lemma J.1 (Per-output-pixel gain identity). *For D_s defined in Eq. 3 and white noise $\eta \sim \mathcal{N}(0, \sigma^2 I)$,*

$$\gamma_{\downarrow}(s) = \|K_{g(s)}\|_F^2.$$

Proof. Stationarity of white noise and Eq. 22 give

$$\mathbb{E}[\|K_{g(s)} * \eta\|_2^2] = (HW)\sigma^2 \|K_{g(s)}\|_F^2.$$

Downsampling by s keeps every s -th sample along each axis: The retained samples all have equal variance as the original, pre-downsampled field. Therefore,

$$\mathbb{E}[\|D_s \eta\|_2^2] = \frac{HW}{s^2} \sigma^2 \|K_{g(s)}\|_F^2,$$

and the normalization in Eq. 25 yields $\gamma_{\downarrow}(s) = \|K_{g(s)}\|_F^2$. $\square$

Proof of Theorem 4.3. By Lemma J.1, $\gamma_{\downarrow}(s) = \|K_{g(s)}\|_F^2$. Applying Theorem 4.1 with kernel size $k = g(s)$ gives

$$\gamma_{\downarrow}(s) \leq \frac{C}{g(s)^2} \leq \frac{C}{(c_1 s)^2} = \frac{C'}{s^2},$$

with $C' = C/c_1^2$ independent of s . $\square$

K. Proof of Theorem 5.1 (Average and Max Poolings under Gaussian Noise)

Consider a pooling window of size $k \geq 2$ in a single channel. Let the clean activations be $S = (S_1, \dots, S_k) \in \mathbb{R}^k$ and let the observation be $S + \eta$, where $\eta = (\eta_1, \dots, \eta_k) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. We define

$$X_{\text{avg}} := \frac{1}{k} \sum_{i=1}^k (S_i + \eta_i), \quad X_{\text{max}} := \max_{1 \leq i \leq k} (S_i + \eta_i),$$

and their clean counterparts $S_{\text{avg}} = \frac{1}{k} \sum_i S_i$, $S_{\text{max}} = \max_i S_i$. Let the errors be $\delta_{\text{avg}} := X_{\text{avg}} - S_{\text{avg}}$, $\delta_{\text{max}} := X_{\text{max}} - S_{\text{max}}$. Write the order statistics $S_{(1)} \geq \dots \geq S_{(k)}$, define the gap $\Delta := S_{(1)} - S_{(2)} \geq 0$, and the standardized gap $z := \Delta/\sigma$. We use $Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $M_k := \max_{1 \leq i \leq k} Z_i$, and $A_k := \max_{1 \leq i \leq k} |Z_i|$.

Proof of (i). By definition, $\delta_{\text{avg}} = \frac{1}{k} \sum_{i=1}^k \eta_i$. Hence

$$\mathbb{E}[\delta_{\text{avg}}] = \frac{1}{k} \sum_i \mathbb{E}[\eta_i] = 0, \quad \text{Var}[\delta_{\text{avg}}] = \frac{1}{k^2} \sum_i \text{Var}[\eta_i] = \frac{\sigma^2}{k}.$$

This part requires only i.i.d. zero-mean noise with variance σ^2 . $\square$

Proof of (ii). (Positive bias) Let $i^* \in \arg \max_i S_i$. Then $X_{\text{max}} \geq S_{i^*} + \eta_{i^*}$. Taking expectations and using $\mathbb{E}[\eta_{i^*}] = 0$ yields

$$\mathbb{E}[\delta_{\text{max}}] = \mathbb{E}[X_{\text{max}} - S_{\text{max}}] \geq 0.$$

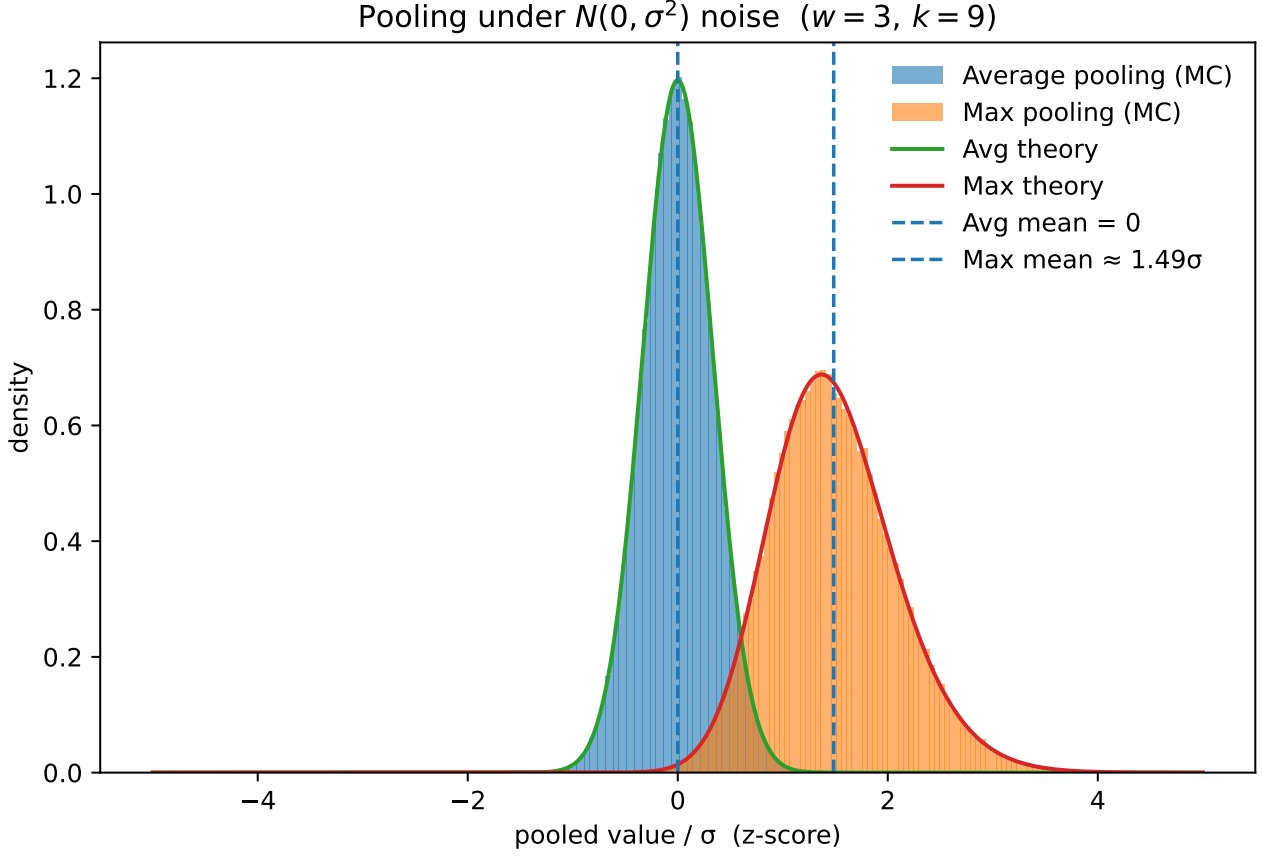


Figure 8. Illustration of positive bias introduced by max pooling

(Uniform-signal case) If $S_1 = \dots = S_k$, translate so $S_i \equiv 0$. Then $\delta_{\text{max}} = \max_i \eta_i = \sigma M_k$ and

$$\mathbb{E}[\delta_{\text{max}}^2] = \sigma^2 \mathbb{E}[M_k^2].$$

Classical Gaussian extreme-value asymptotics (Hall, 1979) give

$$\mathbb{E}[M_k] = \sqrt{2 \log k} - \frac{\log \log k + \log(4\pi)}{2\sqrt{2 \log k}} + o((\log k)^{-1/2}), \quad \text{Var}[M_k] = \frac{\pi^2}{12 \log k} + o((\log k)^{-1}),$$

hence

$$\mathbb{E}[M_k^2] = \text{Var}[M_k] + (\mathbb{E}[M_k])^2 = 2 \log k - \log \log k - \log(4\pi) + o(1),$$

Because $\delta_{\text{max}} = \sigma M_k$, we have

$$\mathbb{E}[\delta_{\text{max}}^2] = \sigma^2 \mathbb{E}[M_k^2] = \sigma^2 (2 \log k - \log \log k - \log(4\pi) + o(1)) = \Theta(\sigma^2 \log k),$$

so the MSE scales as $\Theta(\sigma^2 \log k)$.

(General case) For any realization,

$$|\delta_{\max}| = \left| \max_i (S_i + \eta_i) - \max_i S_i \right| \leq \max_i |\eta_i| = \sigma A_k.$$

Hence

$$\mathbb{E}[\delta_{\max}^2] \leq \sigma^2 \mathbb{E}[A_k^2].$$

We now bound $\mathbb{E}[A_k^2]$ explicitly.

Lemma K.1. For $A_k = \max_{1 \leq i \leq k} |Z_i|$ with $Z_i \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$, we have $\mathbb{E}[A_k^2] \leq 2 \log(2k) + 2$.

Proof of Lemma K.1. For $t \geq 0$, $\Pr(A_k \geq t) \leq \sum_{i=1}^k \Pr(|Z_i| \geq t) \leq 2ke^{-t^2/2}$, where the last step uses the union bound and the standard Gaussian tail estimate $\Pr(|Z| \geq t) \leq 2e^{-t^2/2}$ for $Z \sim \mathcal{N}(0, 1)$; see, e.g., Vershynin (2018). Using $\mathbb{E}[X^2] = \int_0^\infty 2t \Pr(X \geq t) dt$ for a nonnegative X and splitting at $t_0 := \sqrt{2 \log(2k)}$,

$$\begin{aligned} \mathbb{E}[A_k^2] &= \int_0^{t_0} 2t \Pr(A_k \geq t) dt + \int_{t_0}^\infty 2t \Pr(A_k \geq t) dt \\ &\leq t_0^2 + \int_{t_0}^\infty 4kte^{-t^2/2} dt = 2 \log(2k) + 4ke^{-t_0^2/2}. \end{aligned}$$

Because $e^{-t_0^2/2} = e^{-\log(2k)} = 1/(2k)$, the last term equals 2, proving the claim. $\square$

By Lemma K.1,

$$\mathbb{E}[\delta_{\max}^2] \leq \sigma^2 (2 \log(2k) + 2).$$

$\square$

Proof of (iii). Let $T_{\text{avg}}(n) = \frac{1}{k} \sum_i n_i$. By Cauchy-Schwarz, $|T_{\text{avg}}(n)| \leq \|n\|_2 \|k^{-1}(1, \dots, 1)\|_2 = \|n\|_2 / \sqrt{k}$, so $\|T_{\text{avg}}\|_{\ell_2 \rightarrow \ell_2} = k^{-1/2}$, tight for constant n . For max, for any a, b , $|\max_i a_i - \max_i b_i| \leq \|a - b\|_\infty \leq \|a - b\|_2$, hence $\|T_{\text{max}}\|_{\ell_2 \rightarrow \ell_2} \leq 1$, tight for a one-hot n . $\square$

Proof of (iv). Translate so $S_{(1)} = 0$ and $S_i \leq -\Delta$ for $i \geq 2$. Let $\mathcal{S}$ be the switch event that some $j \geq 2$ overtakes the top index after noise:

$$\mathcal{S} := \{\exists j \geq 2 : \eta_j - \Delta \geq \eta_1\} = \{\exists j \geq 2 : Z_j - Z_1 \geq z\}.$$

Because Z_j and Z_1 are independent standard normals, we have $Z_j - Z_1 \sim \mathcal{N}(0, 2)$; hence, by a union bound $\Pr(\mathcal{S}) \leq (k-1) \Pr(\mathcal{N}(0, 2) \geq z) \leq (k-1) e^{-z^2/4} \rightarrow 0$ as $z \rightarrow \infty$. On $\mathcal{S}^c$, $X_{\max} = S_{(1)} + \eta_1 = \eta_1$, so $\delta_{\max}^2 = \eta_1^2$. Dominated convergence then gives $\mathbb{E}[\delta_{\max}^2] \rightarrow \mathbb{E}[\eta_1^2] = \sigma^2$. $\square$

L. Empirical Simulations for Testing Assumption and Theorems

We performed module-level simulations to compare empirical results with the expected values stated in the assumption and theorems. All simulation results closely matched the theoretical expectations. The Python source code used for these simulations is available at https://github.com/kmbmjn/gaussian_noise_robustness.

A_{roll} We embed each $k \times k$ kernel into a 512×512 grid, compute the normalized spectrum $|\widehat{K}|$, form its ℓ_2 -radial profile, and fit the low-pass envelope $\phi_k(r) = (1 + \beta kr)^{-(1+\delta)}$ by weighted log-MSE (Table 22). For representative radii of $\pi/8, \pi/4, \pi/2$, we observed that the empirical magnitudes lie below the fitted envelopes, which verifies this assumption in practice.

Theorem 4.1 Table 23 reports the Monte Carlo estimate of the per-pixel noise gain γ for a $k \times k$ normalized box filter. We convolve i.i.d. $\mathcal{N}(0, \sigma^2)$ noise with the filter via FFT-based circular convolution and compare the empirical $\hat{\gamma}$ with the theoretical $\|K_k\|_F^2 = 1/k^2$, where K_k is the normalized $k \times k$ box stem kernel.

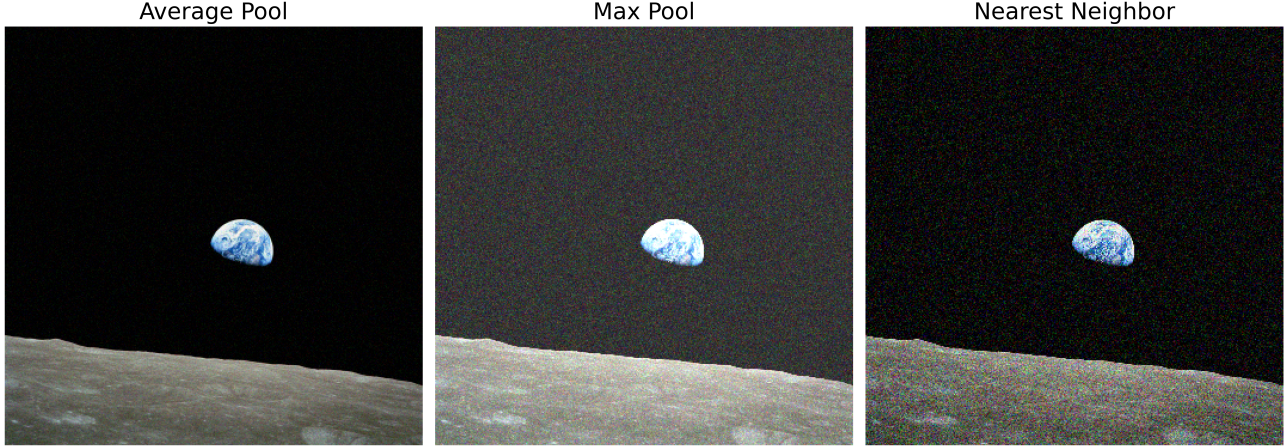


Figure 9. Examples of pooling outputs from a noisy image using average, max, and nearest neighbor

Table 22. The upper block reports the results for the box kernel. The lower block reports the results for the Gaussian kernel.

r (rad)	Empirical ($ \hat{K}_k(\omega) $)	Theoretical ($\phi_k(\ \omega\)$)
0.3962	0.0297134	0.0570019
0.7886	0.0129235	0.0167515
1.5661	0.0040941	0.0042482
0.3962	0.0226295	0.0326660
0.7886	0.0059380	0.0068950
1.6031	0.0007060	0.0010978

Table 23. Measured γ for a $k \times k$ kernel. Stds for 100 simulations are reported.

k	Empirical	Theoretical
4	0.062535 ± 0.001019	0.062500
8	0.015584 ± 0.000484	0.015625
12	0.006911 ± 0.000296	0.006944
16	0.003888 ± 0.000212	0.003906
20	0.002487 ± 0.000169	0.002500
24	0.001728 ± 0.000148	0.001736
28	0.001271 ± 0.000129	0.001276
32	0.000973 ± 0.000115	0.000977

Theorem 4.3 Table 24 reports Monte Carlo estimates of the per-output-pixel noise gain $\gamma_{\downarrow}(s)$ under anti-aliased down-sampling by a factor s , using a $g(s) \times g(s)$ normalized box prefilter and decimation. We compare the empirical $\hat{\gamma}_{\downarrow}$ with the theoretical $\|K_{g(s)}\|_F^2 = 1/g(s)^2$, implying $\sim s^{-2}$ when $g(s) \propto s$.

Theorem 5.1 The results in Table 25 were obtained via Monte Carlo with 200k trials on $S + \eta$ with $\eta \sim \mathcal{N}(0, 1)$ and $k = w^2$. Theoretical entries correspond to σ^2/k for average pooling and Gauss-Hermite quadrature for $E[M_k]$ and $E[M_k^2]$ to compute max-pooling bias and MSE.

Theorem 6.1 We construct random linear maps A with $\|A\|_2 = L_z = 3.0$, compose them with $D = \text{diag}(1/\sigma)$ from INCEPTION and OPENAI, and estimate $\|AD\|_2$ via power iteration. Table 26 compares the theoretical bound $L_z/\sigma_{\min}$ with the measured norm and their ratio, confirming the predicted $1/\sigma_{\min}$ scaling.

Table 24. Measured $\gamma_{\downarrow}(s)$ for anti-aliased downsampling by a factor of s . Stds for 100 simulations are reported.

s	Empirical	Theoretical
1	0.999667 ± 0.006170	1.000000
2	0.250114 ± 0.002767	0.250000
3	0.110970 ± 0.001869	0.111111
4	0.062447 ± 0.001483	0.062500
6	0.027772 ± 0.000880	0.027778
8	0.015567 ± 0.000649	0.015625
12	0.006945 ± 0.000433	0.006944
16	0.003925 ± 0.000359	0.003906

Table 25. Comparison of empirical (Em.) and theoretical (Th.) results for average and max poolings

w	k	Avg MSE (Em.)	Avg MSE (Th.)	Max Bias (Em.)	Max Bias (Th.)	Max MSE (Em.)	Max MSE (Th.)
2	4	0.25083	0.25000	1.02936	1.02938	1.55372	1.55133
3	9	0.11049	0.11111	1.48535	1.48501	2.56409	2.56262
4	16	0.06265	0.06250	1.76524	1.76599	3.41148	3.41374
5	25	0.04006	0.04000	1.96619	1.96531	4.12369	4.12097
6	36	0.02779	0.02778	2.11722	2.11812	4.71818	4.72069

 Table 26. Measured $\|AD\|_2$ closely matches the bound $L_z/\sigma_{\min}$ for random A under INCEPTION and OPENAI, confirming the $1/\sigma_{\min}$ scaling

Constants	Bound $L_z/\sigma_{\min}$	Measured $\ AD\ _2$	$\frac{L_z/\sigma_{\min}}{\ AD\ _2}$
INCEPTION	6.000000	5.998213	1.000298
OPENAI	11.480943	11.200055	1.025079